Technische Universität Bergakademie Freiberg

Faculty of Mechanical, Process and Energy Engineering

Institute of Mechanics and Fluid Dynamics

Chair of Applied Mechanics – Solid Mechanics

Application of Built-in Adaptive Remeshing and Mesh Refinement Features in Abaqus to Fracture Simulations Using Phase-field User Elements

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Author: Pruthviraja Reddy Vandavagali

Matriculation number: 68865

1st Examiner (Supervisor): Prof. Dipl.-Ing. Björn Kiefer, Ph.D.

2nd Examiner (Reviewer): Dr.-Ing. Stephan Roth

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Abstract

Phase-field fracture methods implemented within commercial finite element software via custom user subroutines (UEL/UMAT) provide a powerful variational framework for regularized crack initiation, propagation, and complex branching. However, resolving the regularizing material length scale l_0 requires extremely fine spatial discretizations, rendering uniform fine meshes computationally prohibitive for large-scale engineering structures. This thesis investigates the application of error-indicator-guided mesh refinement, multifaceted numerical convergence, and nonmatching multi-field state transfer to phase-field brittle fracture simulations in Abaqus/Standard.

The investigation is established upon the canonical Mode-I single-edge-notched tensile fracture benchmark. First, the boundary value problem is formulated using a three-layer co-located element architecture in user subroutine `f42_mixed_uel.for`, rigorously disambiguating material modulus E from structural compliance K_0 and demonstrating that a zero-gap duplicate-node seam is essential to prevent artificial notch compliance. Unconstrained linear regression establishes the canonical S_1 baseline (15,192 elements, Job `1406015.mmaster02`, $K_0 = 137.945520\,\mathrm{kN/mm}$, $F_{\max} = 0.757778\,\mathrm{kN}$) as the quantitative anchor for all subsequent evaluations.

To alleviate the severe computational cost of uniform grids, an error-indicator-guided pre-refinement methodology is implemented following Pandey and Kumar [21]. A linear elastic pre-analysis extracts the recovery-based Von Mises stress discretization error indicator (`MISESERI`) based on the Zienkiewicz–Zhu energy norm. Investigation of preprocessor execution revealed that standard Abaqus `*NSET` keyword card limits silently discarded boundary nodes, introducing edge lift and compliance loss; multi-line card wrapping resolved this anomaly and restored reference-consistent stiffness within 0.09% ($K_0 = 137.820804\,\mathrm{kN/mm}$). Graded meshes concentrate fine resolution ($h \leq l_0/2$) exclusively along the prospective crack corridor.

To establish the prerequisites for multi-stage adaptive simulation, a multi-field state-transfer framework is formulated across nonmatching discretizations. The transfer pipeline combines shape-function interpolation for primary kinematic and damage degrees of freedom with centroid-based mapping and strict positivity clamping for the internal crack-driving history variable $\mathcal{H}$. State transfer is systematically verified across same-mesh restarts, multi-resolution refined and coarsened grids, and a synthetic single-facet crack-tip topology cut with post-peak continuation under a four-stage restart protocol. These studies establish kinematic and state-admissibility qualification for the tested configurations, while full energetic preservation during transfer is explicitly identified as pending Gate 6C qualification.

A multifaceted convergence evaluation demonstrates that numerical adequacy cannot be assessed by peak reaction force alone. Across spatial resolutions (S_1–S_4, $h/l_0 \in [0.167, 0.400]$)

and historical fine-mesh data, structural stiffness K_0 exhibits exceptional invariance ($\Delta < 0.09\%$), while peak force exhibits monotonic mesh sensitivity (4.26% reduction). Temporal incrementation (T_1–T_3, $0.5\times$ to $2.0 \times \Delta t$) confirms mechanical peak load invariance ($\Delta < 0.07\%$) alongside energetic temporal sensitivity. A controlled three-point physical length-scale study on a fixed S_3 grid ($l_0 \in \{7.5, 11.25, 15.0\}\,\mu$m) demonstrates stiffness stability (0.13%) and observed limit-load reductions (5.83%). In adaptive meshes (A_1–A_4), post-peak softening resistance is characterized under the crack-pinning hypothesis (`OBSERVED_FORCE_PLATEAU`).

Finally, a comprehensive UEL energy audit is implemented. A Single-IP extraction rule eliminates a $4\times$ visualization overcounting artifact and recovers the once-per-underlying-finite-element global energy sum under the 2D implicit-unit-thickness convention. Mechanical parity against uninstrumented solvers is rigorously qualified. Because standard Abaqus output databases do not record within-increment Newton paths and cross-derivatives do not commute for the implemented staggered displacement–damage coupling, the endpoint bookkeeping difference remains within $\pm 0.008\%$ pre-peak ($+0.7607\%$ post-peak for S_1/T_2), and `GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED` is preserved. This thesis provides an evidence-based foundation and practical decision tree for error-guided refinement and state transfer in phase-field fracture mechanics.

Chapter 1

Introduction, Theoretical Foundations, and Literature Review

1.1 Motivation and Problem Statement

Predicting crack initiation, propagation, and complex branching patterns in brittle and quasi-brittle solids is a central challenge in computational solid mechanics and materials engineering [4]. Classical linear elastic fracture mechanics (LEFM), initiated by Griffith [12] and extended by Irwin, provides rigorous criteria for crack growth based on energy release rates and stress intensity factors. However, standard discrete fracture models—such as cohesive zone models or the extended finite element method (XFEM)—require explicit tracking of geometric discontinuities, geometric enrichment, or complicated remeshing procedures when handling arbitrary three-dimensional crack paths, crack coalescence, or spontaneous branching.

The **variational phase-field approach to fracture**, pioneered by Francfort and Marigo [11] and computationally realized by Bourdin, Francfort, and Marigo [5, 6], overcomes these difficulties by regularizing sharp crack surfaces into a continuous diffusive damage corridor governed by a continuous scalar field $d \in [0, 1]$. In this framework, crack topology is implicitly tracked by the phase field, and the crack path emerges naturally from energy minimization without requiring ad-hoc tracking criteria or discontinuous displacement jumps [15].

Despite its mathematical elegance, the phase-field method imposes severe computational demands. To accurately capture the steep gradients of the phase field across the regularized fracture surface, the finite element mesh size h must sufficiently resolve the internal material length scale l_0. Standard convergence studies dictate that at least two to five elements must span the half-width of the diffuse zone ($h \leq l_0/2$). For fine structural features or thin specimens where l_0 is on the order of micrometers, a uniform discretization of the entire domain leads to massive system dimensions (10^5 to 10^7 degrees of freedom), making multi-step or full post-peak fracture simulations computationally prohibitive [13, 22].

To mitigate this computational bottleneck, adaptive mesh refinement strategies have become essential [3, 20]. In commercial finite element suites such as Abaqus, user-defined element subroutines (UELs) and user material subroutines (UMATs) are commonly employed to solve coupled phase-field systems [18, 19, 8]. However, integrating Abaqus' built-in adaptive remeshing

tools with multi-field UEL formulations presents non-trivial challenges:

1. Standard built-in error indicators (such as the recovery-based stress error indicator `MISESERI`) operate primarily on native continuum elements rather than custom multi-field user elements.

2. Commercial solvers lack native state-mapping routines for custom UEL integration-point history variables ($\mathcal{H}$) across nonmatching grids.

3. Dynamic crack growth requires rigorous preservation of thermodynamic bounds ($0 \leq d \leq 1$) and irreversible crack growth ($\dot{d} \geq 0$) when transferring state data between meshes.

This thesis addresses these challenges by investigating the application of Abaqus built-in error indicators, error-guided local mesh refinement, and nonmatching multi-field state transfer to phase-field fracture simulations.

1.2 Variational Formulation of Phase-Field Brittle Fracture

1.2.1 Regularized Energy Functional

Consider an elastic body occupying a domain $\Omega \subset \mathbb{R}^\delta$ ($\delta \in \{2, 3\}$) with external boundary $\partial\Omega = \Gamma_u \cup \Gamma_t$ ($\Gamma_u \cap \Gamma_t = \emptyset$). Following the variational formulation of Francfort and Marigo [11] and Bourdin et al. [5], the total potential energy of the fracturing solid is expressed as the sum of the elastic strain energy and the fracture surface energy:

$$\Pi(\mathbf{u}, d) = \int_\Omega \psi(\boldsymbol{\varepsilon}(\mathbf{u}), d)\, \mathrm{d}\Omega + G_c \int_\Omega \gamma(d, \nabla d)\, \mathrm{d}\Omega - \int_\Omega \mathbf{b} \cdot \mathbf{u}\, \mathrm{d}\Omega - \int_{\Gamma_t} \mathbf{t} \cdot \mathbf{u}\, \mathrm{d}\Gamma, \tag{1.1}$$

where $\mathbf{u}(\mathbf{x})$ is the displacement field, $\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla\mathbf{u} + (\nabla\mathbf{u})^\mathsf{T})$ is the small-strain tensor, G_c is the critical energy release rate (fracture toughness), $\mathbf{b}$ denotes body forces, $\mathbf{t}$ represents prescribed surface tractions, and $\gamma(d, \nabla d)$ is the crack surface density function.

The continuous scalar phase field $d(\mathbf{x}, t) \in [0, 1]$ serves as an interpolation parameter between the undamaged elastic state ($d = 0$) and the completely broken fracture state ($d = 1$). In this work, we adopt the standard second-order regularized AT2 crack surface density function [5, 16]:

$$\gamma(d, \nabla d) = \frac{1}{2l_0} d^2 + \frac{l_0}{2} |\nabla d|^2, \tag{1.2}$$

where $l_0 > 0$ is the regularization parameter possessing the dimension of length, governing the spatial width of the diffuse crack band. In the asymptotic limit $l_0 \to 0$, the regularized crack surface energy $\int_\Omega G_c \gamma(d, \nabla d)\, \mathrm{d}\Omega$ Γ-converges to the Griffith discrete crack surface energy $G_c \mathcal{H}^{\delta-1}(\Gamma_c)$ [6].

1.2.2 Strain Energy Decomposition and Constitutive Degradation

To prevent crack propagation and unphysical material interpenetration under compressive stress states, the undamaged elastic strain energy density $\psi_0(\boldsymbol{\varepsilon})$ is decomposed in foundational variational formulations into positive (tensile) and negative (compressive) parts:

$$\psi_0(\boldsymbol{\varepsilon}) = \psi_+(\boldsymbol{\varepsilon}) + \psi_-(\boldsymbol{\varepsilon}). \tag{1.3}$$

In the spectral strain decomposition proposed by Miehe, Welschinger, and Hofacker [16], the energy split is formulated as:

$$\psi_+(\boldsymbol{\varepsilon}) = \frac{1}{2}\lambda\langle \mathrm{tr}(\boldsymbol{\varepsilon})\rangle_+^2 + \mu\,\mathrm{tr}\left(\boldsymbol{\varepsilon}_+^2\right), \tag{1.4}$$

$$\psi_-(\boldsymbol{\varepsilon}) = \frac{1}{2}\lambda\langle \mathrm{tr}(\boldsymbol{\varepsilon})\rangle_-^2 + \mu\,\mathrm{tr}\left(\boldsymbol{\varepsilon}_-^2\right), \tag{1.5}$$

where λ and μ are the standard Lamé elastic constants, $\langle x\rangle_\pm = \frac{1}{2}(x \pm |x|)$ denote Macaulay brackets, and the positive/negative strain tensors are constructed from the spectral decomposition of the strain tensor $\boldsymbol{\varepsilon} = \sum_{i=1}^{\delta} \varepsilon_i \mathbf{n}_i \otimes \mathbf{n}_i$:

$$\boldsymbol{\varepsilon}_+ = \sum_{i=1}^{\delta} \langle \varepsilon_i\rangle_+ \mathbf{n}_i \otimes \mathbf{n}_i, \qquad \boldsymbol{\varepsilon}_- = \sum_{i=1}^{\delta} \langle \varepsilon_i\rangle_- \mathbf{n}_i \otimes \mathbf{n}_i. \tag{1.6}$$

In standard Miehe theory, constitutive damage degradation acts exclusively on the tensile strain energy component $\psi_+(\boldsymbol{\varepsilon})$ through a monotonic quadratic degradation function $g(d)$:

$$\psi_{\mathrm{Miehe}}(\boldsymbol{\varepsilon}, d) = g(d)\psi_+(\boldsymbol{\varepsilon}) + \psi_-(\boldsymbol{\varepsilon}) = \left[(1-d)^2 + k_{\mathrm{res}}\right]\psi_+(\boldsymbol{\varepsilon}) + \psi_-(\boldsymbol{\varepsilon}), \tag{1.7}$$

where $k_{\mathrm{res}} \ll 1$ (typically $k_{\mathrm{res}} = 10^{-7}$) is a positive residual numerical stiffness parameter that ensures positive-definiteness of the mechanical tangent operator in fully broken regions ($d \to 1$). The theoretical Cauchy stress tensor is then:

$$\boldsymbol{\sigma}_{\mathrm{Miehe}} = \frac{\partial \psi_{\mathrm{Miehe}}}{\partial \boldsymbol{\varepsilon}} = g(d)\boldsymbol{\sigma}_+ + \boldsymbol{\sigma}_-, \tag{1.8}$$

with $\boldsymbol{\sigma}_\pm = \lambda\langle \mathrm{tr}(\boldsymbol{\varepsilon})\rangle_\pm \mathbf{I} + 2\mu\boldsymbol{\varepsilon}_\pm$.

Formulation Implemented in the User Subroutine (`f42_mixed_uel.for`): An important implementation distinction must be drawn between the above general Miehe split theory and the actual constitutive formulation implemented in the governed user element source code (`f42_mixed_uel.for`). To maintain computational efficiency, robust tangent symmetry, and straightforward convergence in staggered Abaqus/Standard analyses, the implemented code employs a **hybrid formulation**:

1. **Tensile Energy for Phase-Field Driving**: The spectral tensile strain energy density $\psi_+(\boldsymbol{\varepsilon})$ of Eq. (1.4) is evaluated at element quadrature points and passed to the crack-driving history variable $\mathcal{H}$. This strictly prevents crack initiation or propagation under compressive stress fields.

2. **Isotropic Degradation in Mechanical Balance**: In the displacement user element (Layer 2), the mechanical stress tensor is evaluated using full isotropic stiffness degradation driven by an element-averaged damage variable $\bar{d}$:

$$\boldsymbol{\sigma} = g(\bar{d})\mathbb{C}_0 : \boldsymbol{\varepsilon} = \left[(1-\bar{d})^2 + k_{\mathrm{res}}\right]\mathbb{C}_0 : \boldsymbol{\varepsilon}, \tag{1.9}$$

   where $\mathbb{C}_0$ is the standard fourth-order isotropic linear-elastic stiffness tensor. The corresponding algorithmic mechanical material Jacobian returned to Abaqus is simply $\mathbb{C}_{\mathrm{alg}} = g(\bar{d})\mathbb{C}_0$.

This hybrid formulation captures the unilateral crack-driving physics of the spectral decomposition while preserving a linear, symmetric mechanical tangent operator at fixed damage.

1.2.3 Irreversibility and the Crack-Driving History Field

Fracture is inherently an irreversible dissipative process, requiring that crack healing is strictly prevented:

$$\dot{d}(\mathbf{x}, t) \geq 0 \quad \forall\, \mathbf{x} \in \Omega,\ t \geq 0. \tag{1.10}$$

To enforce irreversibility without introducing computationally expensive constrained optimization or penalty projections at the global degree-of-freedom level, Miehe, Hofacker, and Welschinger [17] introduced a local historical maximum tensile strain energy density field $\mathcal{H}(\mathbf{x}, t)$:

$$\mathcal{H}(\mathbf{x}, t) = \max_{\tau \in [0,t]} \psi_+(\boldsymbol{\varepsilon}(\mathbf{x}, \tau)), \tag{1.11}$$

satisfying the initial condition $\mathcal{H}(\mathbf{x}, 0) = 0$ and the local update rule $\mathcal{H}_{n+1} = \max(\mathcal{H}_n, \psi_+(\boldsymbol{\varepsilon}_{n+1}))$.

Replacing $\psi_+(\boldsymbol{\varepsilon})$ by $\mathcal{H}(\mathbf{x}, t)$ in the phase-field driving variational derivative yields the modified phase-field governing Euler–Lagrange equation:

$$d - l_0^2 \nabla^2 d = \frac{2l_0}{G_c}(1 - d)\mathcal{H} \quad \text{in } \Omega, \tag{1.12}$$

subject to the homogeneous Neumann boundary condition $\nabla d \cdot \mathbf{n} = 0$ on $\partial\Omega$.

1.2.4 Coupled Weak Form and Staggered Solution Algorithm

Multiplying by test functions $\delta\mathbf{u} \in \mathcal{V}_u = \{\mathbf{v} \in [H^1(\Omega)]^\delta : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_u\}$ and $\delta d \in \mathcal{V}_d = H^1(\Omega)$, the coupled weak system governing momentum balance and phase-field evolution in the implemented hybrid scheme reads:

$$\int_\Omega g(\bar{d})(\mathbb{C}_0 : \boldsymbol{\varepsilon}) : \boldsymbol{\varepsilon}(\delta\mathbf{u})\, \mathrm{d}\Omega = \int_\Omega \mathbf{b} \cdot \delta\mathbf{u}\, \mathrm{d}\Omega + \int_{\Gamma_t} \mathbf{t} \cdot \delta\mathbf{u}\, \mathrm{d}\Gamma, \tag{1.13}$$

$$\int_\Omega \left[\left(\frac{G_c}{l_0} + 2\mathcal{H} \right) d\, \delta d + G_c l_0 \nabla d \cdot \nabla(\delta d) \right] \mathrm{d}\Omega = \int_\Omega 2\mathcal{H}\, \delta d\, \mathrm{d}\Omega. \tag{1.14}$$

Due to the non-convex coupling between $\mathbf{u}$ and d, monolithic Newton–Raphson schemes often suffer from severe convergence difficulties and loss of positive-definiteness during softening [2, 10]. In this work, we utilize a robust **staggered (operator-split) solution algorithm** [5, 17, 18]:

1. At load increment t_{n+1}, freeze the phase field at d_n and solve the mechanical equilibrium Eq. (1.13) for displacement $\mathbf{u}_{n+1}$.
2. Compute the local tensile energy $\psi_+(\boldsymbol{\varepsilon}_{n+1})$ at all element integration points and update the history field $\mathcal{H}_{n+1} = \max(\mathcal{H}_n, \psi_+)$.
3. Freeze the history field at $\mathcal{H}_{n+1}$ and solve the strictly linear, symmetric, and positive-definite phase-field subproblem Eq. (1.14) for d_{n+1}.

1.3 Abaqus User-Element (UEL/UMAT) Architecture

Commercial finite element suites like Abaqus/Standard do not provide native elements with coupled displacement and phase-field degrees of freedom. Consequently, the multi-field system is implemented through user-defined subroutines [18, 19, 8].

In the implementation utilized throughout this thesis (Fig. 1.1):

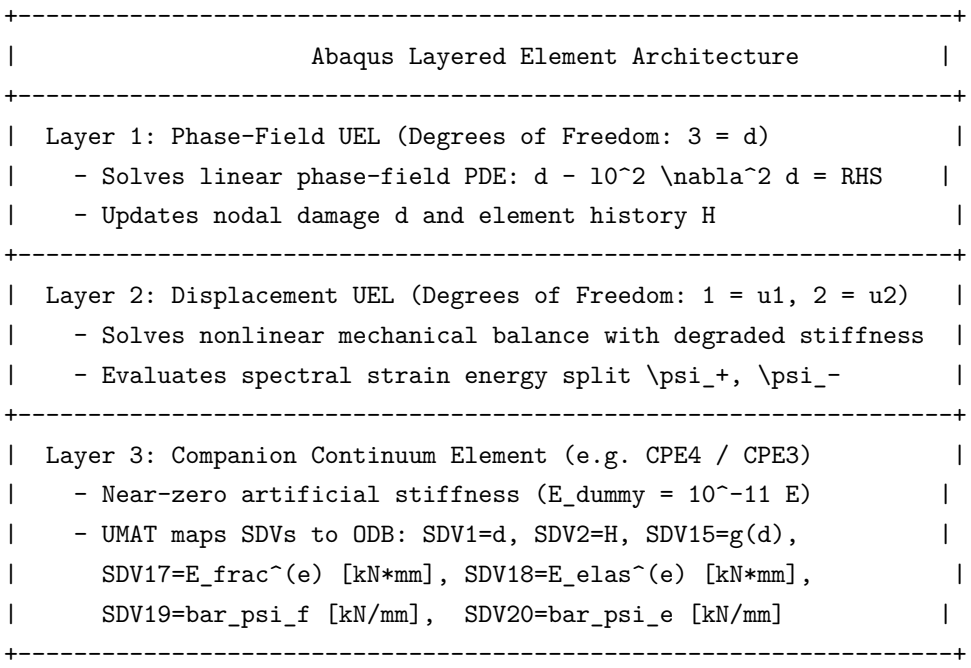


Figure 1.1: Schematic representation of the layered user-element (UEL/UMAT) architecture in Abaqus.

- **Physical Mesh vs Layered Mesh**: For every physical geometric quad or triangle element, multiple coincident element layers are generated sharing identical nodal coordinates.
- **Phase-Field UEL Layer**: Elements assigned to user element type `U1` (`JTYPE = 1` quad, `JTYPE = 3` tri) solve the Helmholtz-type phase equation. In 2D plane-strain models, the scalar damage unknown d is mapped to Abaqus degree of freedom 3 (`DOF 3`, variable `U3`) on co-located nodes, utilizing the out-of-plane displacement component without requiring thermal element types.
- **Mechanical Displacement UEL Layer**: Elements assigned to user element type `U2` (`JTYPE = 2` quad, `JTYPE = 4` tri) solve the degraded elasticity equation with active in-plane displacement degrees of freedom `DOF 1` (u_1) and `DOF 2` (u_2).
- **Visualization / Companion Layer**: Standard Abaqus continuum elements (`CPE4`, `CPE3`) with near-zero artificial stiffness ($E_{\text{dummy}} \approx 10^{-11}E$) are superposed over the physical domain. User subroutine `UMAT` copies integration-point state variables (SDVs) into this layer, enabling standard Abaqus/Viewer field rendering:
  - `SDV1` (mirrored to `SDV14`): Scalar damage phase field $d \in [0, 1]$.
  - `SDV2` (mirrored to `SDV16`): Internal crack-driving history variable $\mathcal{H}$.
  - `SDV15`: Degraded stiffness factor $g(d) = (1 - d)^2 + k_{\text{res}}$ (with residual parameter $k_{\text{res}} = 10^{-7}$).
  - `SDV17`: Whole-underlying-finite-element fracture surface energy $E_{\text{frac}}^{(e)}$ in native source units $\text{kN} \cdot \text{mm} = \text{J}$ (converted to mJ in reporting via a single $1000\times$ multiplication

factor; not natively stored in mJ).

- `SDV18`: Whole-underlying-finite-element elastic strain energy $E^{(e)}_{\text{elas}}$ in native source units $\text{kN} \cdot \text{mm} = \text{J}$ (converted to mJ in reporting via a single $1000\times$ multiplication factor; not natively stored in mJ).
- `SDV19` / `SDV20`: Area-normalized source quantities $\bar{\psi}_f$ and $\bar{\psi}_e$ with direct source-algebra dimension $\text{kN/mm} = \text{J/mm}^2$ ($E^{(e)}/A_e$); interpretable as volumetric energy densities ($\text{kN/mm}^2 = 1\,\text{J/mm}^3 = 1000\,\text{mJ/mm}^3$) only under the explicit 1-mm implicit-thickness convention ($B = 1\,\text{mm}$).

- **Inter-Subroutine Shared State**: Common blocks and global arrays accessed via `UEXTERNALDB(LOP=0)` communicate committed history $\mathcal{H}$ and phase fields between staggered steps and element layers.

1.4 Error Estimation and Mesh Refinement Principles

1.4.1 Zienkiewicz–Zhu Recovery-Based Error Estimator

To guide mesh refinement toward regions requiring fine spatial resolution without prior knowledge of the crack trajectory, a posteriori error estimators are employed [3]. Following Zienkiewicz and Zhu [25, 26], the spatial discretization error $\mathbf{e}_\sigma = \boldsymbol{\sigma}^* - \boldsymbol{\sigma}_h$ is measured by comparing the discontinuous raw finite element stress field $\boldsymbol{\sigma}_h$ with a continuous, recovered superconvergent stress field $\boldsymbol{\sigma}^*$.

The global energy norm of the discretization error is defined as:

$$\|\mathbf{e}_\sigma\|_E = \left(\int_\Omega (\boldsymbol{\sigma}^* - \boldsymbol{\sigma}_h) : \mathbf{C}^{-1} : (\boldsymbol{\sigma}^* - \boldsymbol{\sigma}_h) \, \mathrm{d}\Omega \right)^{1/2} = \left(\sum_{e=1}^{N_{\text{elem}}} \|\mathbf{e}_\sigma\|^2_{E,e} \right)^{1/2}, \tag{1.15}$$

where $\mathbf{C}$ is the fourth-order elasticity tensor and $\|\mathbf{e}_\sigma\|_{E,e}$ is the element-level error contribution. The relative energy norm error η is normalized by the total strain energy U:

$$\eta = \frac{\|\mathbf{e}_\sigma\|_E}{\left(U + \|\mathbf{e}_\sigma\|^2_E\right)^{1/2}} \times 100\%. \tag{1.16}$$

1.4.2 Abaqus MISESERI Formulation

In Abaqus, the built-in stress error indicator `MISESERI` computes an equivalent Von Mises stress discretization error indicator at element integration points [1]:

$$\text{MISESERI}_e = \left(\frac{1}{V_e} \int_{\Omega_e} \left[q(\boldsymbol{\sigma}^*) - q(\boldsymbol{\sigma}_h) \right]^2 \mathrm{d}\Omega \right)^{1/2}, \tag{1.17}$$

where $q(\boldsymbol{\sigma}) = \sqrt{\frac{3}{2}\mathbf{S} : \mathbf{S}}$ is the equivalent Mises stress, $\mathbf{S} = \boldsymbol{\sigma} - \frac{1}{3}\text{tr}(\boldsymbol{\sigma})\mathbf{I}$ is the stress deviator, and V_e is the element volume. High values of `MISESERI` identify elements where spatial stress gradients are poorly resolved by the current mesh polynomial order and element size h_e.

1.4.3 The Pandey–Kumar Pre-Refinement Methodology

Pandey and Kumar [21] developed a simple, robust implementation for phase-field fracture modeling in Abaqus by decoupling error-indicator extraction from the non-linear fracture simulation:

1. **Linear Elastic Pre-Analysis**: Execute a rapid, coarse-grid elastic pre-analysis up to a small threshold displacement prior to damage initiation ($u \ll u_{\text{init}}$).
2. **Error Indicator Extraction**: Extract the resulting spatial distribution of `MISESERI` from the pre-analysis output database (`.odb`).
3. **Remeshing Rule Application**: Construct an Abaqus `RemeshingRule` targeting a designated solution error (e.g., target error 5%) with specified minimum element size ($h_{\min} \leq l_0/2$) and maximum size ($h_{\max}$).
4. **Pre-Refined Mesh Generation**: Generate an optimized, graded mesh where fine element sizing is concentrated along the stress-concentration and anticipated crack-initiation corridor.
5. **Production Fracture Analysis**: Execute the full, non-linear phase-field fracture simulation on the pre-refined mesh from the virgin state.

This approach eliminates the algorithmic complexities of continuous in-analysis remeshing while capturing the primary computational savings of adaptive mesh refinement.

1.5 Multi-Field State Transfer across Nonmatching Meshes

When advancing from pre-refinement toward true multi-stage adaptivity or restart-based analysis [24], state variables from an existing donor mesh $\Omega_D = \bigcup_e \Omega_{D,e}$ must be transferred onto a nonmatching target mesh $\Omega_T = \bigcup_k \Omega_{T,k}$.

1.5.1 Primary Field Transfer (Nodal Interpolation)

Primary field unknowns—namely nodal displacement vectors $\mathbf{u} = (u_1, u_2)^{\mathsf{T}}$ and nodal phase field d—are defined at mesh vertices. For any target node with spatial coordinates $\mathbf{x}_T \in \Omega_T$, geometric spatial search identifies the host donor element $\Omega_{D,e}$ satisfying $\mathbf{x}_T \in \Omega_{D,e}$. The local natural coordinates $(\xi_T, \eta_T) \in [-1, 1]^2$ are inverted via Newton–Raphson:

$$\mathbf{x}_T = \sum_{a=1}^{n_{\text{node}}} N_a(\xi_T, \eta_T)\mathbf{x}_{D,a}, \tag{1.18}$$

and the transferred nodal field $\phi_T(\mathbf{x}_T) \in \{\mathbf{u}_T, d_T\}$ is evaluated directly from donor nodal values $\phi_{D,a}$ [9, 7]:

$$\phi_T(\mathbf{x}_T) = \sum_{a=1}^{n_{\text{node}}} N_a(\xi_T, \eta_T)\phi_{D,a}. \tag{1.19}$$

To preserve thermodynamic admissibility, the transferred phase field is strictly clamped:

$$d_T(\mathbf{x}_T) \leftarrow \min\left(1.0, \max(0.0, d_T(\mathbf{x}_T))\right). \tag{1.20}$$

1.5.2 Secondary History Field Transfer (Gauss-Point Mapping)

Unlike nodal quantities, the crack-driving history variable $\mathcal{H}$ is an internal Gauss-point quantity defined at quadrature points $\mathbf{x}_{T,g}$. Transferring integration-point history variables requires specialized operators [23, 14]:

1. **Direct Quadrature Interpolation**: Evaluate donor Gauss-point values $\mathcal{H}_{D,i}$ via donor element shape functions or nearest-centroid projection.

2. **Positivity and Floor Clamping**: Because $\mathcal{H}$ represents an energy density, negative values are unphysical. The transferred field is strictly projected:

$$\mathcal{H}_T(\mathbf{x}_{T,g}) \geq 0 \quad \forall\, g \in \{1, \ldots, n_{\text{gp}}\}. \tag{1.21}$$

3. **Phase-Consistency Floor (Local AT2 Heuristic)**: To preserve consistency between transferred damage d_T and the internal driving history $\mathcal{H}_T$ without triggering spurious immediate damage growth upon restart, a lower bound is enforced on the transferred history field. In the absence of spatial damage gradients ($\nabla d = \mathbf{0}$), setting the AT2 Euler–Lagrange residual to zero yields the local homogeneous relation:

$$\mathcal{H}_T(\mathbf{x}_{T,g}) \geq \frac{G_c}{2l_0} \frac{d_T(\mathbf{x}_{T,g})}{1 - d_T(\mathbf{x}_{T,g})} \quad \text{for quadrature points with } 0 < d_T < 1. \tag{1.22}$$

   This local algebraic threshold serves as a conservative consistency floor for transferred states rather than a universal Karush–Kuhn–Tucker (KKT) optimality condition for the non-local spatial boundary value problem.

1.6 Scope of Work and Thesis Organization

1.6.1 Distinction of Methodological Scope

To ensure rigorous scientific boundaries, this thesis strictly distinguishes between three related concepts:

- **Offline Error-Guided Pre-Refinement**: Mesh refinement conducted prior to the fracture simulation, guided by linear elastic error indicators (`MISESERI`) at stress concentrators (Chapters 2 and **??**). This approach is fully validated and evaluated.

- **Nonmatching State Transfer & Multi-Grid Restart**: The mathematical and computational framework for mapping primary and history state variables across nonmatching discretizations (Chapters 3 and 4). Kinematic and state-admissibility criteria are systematically verified, while complete energetic preservation remains subject to pending Gate 6C qualification.

- **Online In-Analysis Adaptive Remeshing**: Fully automated, runtime remeshing during crack propagation where the mesh continuously updates and transfers state at every time increment. As documented in Chapters **??** and 7, online dynamic remeshing during crack propagation remains a limitation and future research extension.

1.6.2 Thesis Organization

The remainder of this thesis is structured as follows:

- **Chapter ?? (Mode-I Phase-Field Fracture Benchmark and Reference Baseline)** details the canonical Mode-I single-edge-notched tensile benchmark, disambiguates structural compliance K_0 from material modulus E, verifies sharp crack seam kinematics versus blunt notches, documents the co-located UEL/UMAT architecture, and establishes the canonical S_1 reference baseline ($K_0 = 137.945520\,\text{kN/mm}$, Job `1406015.mmaster02`).
- **Chapter 2 (Error-Indicator-Guided Mesh Refinement Methodology)** investigates Abaqus built-in `MISESERI` stress error extraction, isolates and corrects the preprocessor `*NSET` keyword truncation defect, restores reference stiffness within 0.09%, and generates pre-refined meshes along prospective fracture paths.
- **Chapter 3 (Nonmatching State Transfer and Restart Methodology)** formulates the multi-field transfer operators, shape-function nodal interpolation, Gauss-point history mapping, and four-stage numerical restart protocol.
- **Chapter 4 (Multi-Resolution and Topology-Transfer Benchmarks)** evaluates state transfer across multi-resolution refined and coarsened grids and evaluates a synthetic crack-tip topology separation benchmark.
- **Chapter 5 (High-Performance Computing and Shared-State Architecture)** examines shared-memory thread safety in Abaqus UELs, `UEXTERNALDB` common blocks, cluster scheduler concurrency, and fail-closed dual-channel notification design.
- **Chapter ?? (Multifaceted Mode-I Convergence Evaluation and UEL Energy Audit)** presents comprehensive convergence evaluations across spatial series (S_1–S_4), temporal scaling (T_1–T_3), physical length-scale sensitivity ($l_0 \in \{7.5, 11.25, 15.0\}\,\mu\text{m}$), and adaptive meshes ($A_1$–$A_4$), followed by the UEL energy implementation, Single-IP deduplication rule, mechanical parity, and global energy audit.
- **Chapter 7 (Synthesis, Decision Framework, and Conclusions)** synthesizes findings into a comprehensive epistemological claim ledger, reviews the master milestone gate status, provides an engineering decision tree, and summarizes conclusions.
- **Appendix A (Reproducibility Tables and Execution Ledger)** provides complete HPC scheduler accounting, job IDs, input-deck hashes, and execution provenance logs.

1.7 Problem Formulation and Boundary Conditions

The baseline verification problem represents a square specimen of dimensions $L \times W = 1.0\,\text{mm} \times 1.0\,\text{mm}$ subject to tensile Mode-I loading, adopting the benchmark geometry of Pandey and Kumar [21]. An initial horizontal sharp crack of length $a_0 = 0.5\,\text{mm}$ is embedded along the centerline ($y = 0.5\,\text{mm}$, $0 \leq x \leq 0.5\,\text{mm}$).

Kinematic Boundary Conditions:

- **Bottom Boundary ($y = 0\,\mathbf{mm}$)**: Roller constraint preventing vertical displacement, $u_y(x, 0) = 0$.
- **Bottom-Left Corner Pin ($(x, y) = (0, 0)$)**: Rigid-body horizontal constraint, $u_x(0, 0) = 0$.
- **Top Boundary ($y = 1.0\,\mathbf{mm}$)**: Homogeneous vertical tensile displacement $u_y(x, 1) = u(t)$ prescribed via a reference point (RP) kinematically coupled to the top edge nodes, with free lateral contraction (u_x unconstrained).
- **Lateral Boundaries ($x = 0\,\mathbf{mm}, x = 1.0\,\mathbf{mm}$)**: Traction-free boundaries, $\boldsymbol{\sigma} \cdot \mathbf{n} = \mathbf{0}$.

Constitutive Material and Phase-Field Properties: The material is modeled under plane-strain linear elasticity with AT2 phase-field regularized fracture:

- Young's Modulus: $E = 210\,\mathrm{GPa} \equiv 210.0\,\mathrm{kN/mm^2}$
- Poisson's Ratio: $\nu = 0.3$
- Critical Fracture Energy Release Rate: $G_c = 2.7 \times 10^{-3}\,\mathrm{kN/mm} \equiv 2.7\,\mathrm{N/mm} = 2700\,\mathrm{J/m^2}$
- Phase-Field Regularization Length Scale: $l_0 = 0.0075\,\mathrm{mm} \equiv 7.5\,\mu\mathrm{m}$
- Residual Degradation Parameter: $k_{\mathrm{res}} = 1.0 \times 10^{-7}$

1.8 Crack Geometry Representation: Sharp Seam vs. Blunt Notch

A fundamental mechanical distinction in fracture benchmarks is the representation of initial crack geometry. While literature implementations frequently approximate cracks as finite-width blunt notches ($r_0 > 0$) to avoid singular meshing, the benchmark formulation requires an ideal sharp crack ($a_0 = 0.5\,\mathrm{mm}$) modeled as a zero-gap duplicate-node seam.

Global Structural Stiffness Definition: The initial structural stiffness K_0 is defined as the slope of the linear-elastic reaction force (F) versus prescribed top-edge displacement (u) curve before damage initiation:

$$K_0 = \left.\frac{\mathrm{d}F}{\mathrm{d}u}\right|_{\text{initial elastic range}} \approx \frac{\Delta F}{\Delta u}. \tag{1.23}$$

While Young's modulus E is a local constitutive tensor property ($210\,\mathrm{kN/mm^2}$), K_0 represents the integrated load-deformation compliance of the entire boundary value problem, depending directly on specimen geometry, boundary constraints, and notch kinematics.

Early geometry-sanity-check models (Figure 1.2) compared an uncracked solid ($K_0 \approx 138.15\,\mathrm{kN/mm}$), a blunt notch ($K_0 \approx 75.47\,\mathrm{kN/mm}$), and an early sharp seam diagnostic ($K_0 \approx 141.82\,\mathrm{kN/mm}$), demonstrating that finite-width blunt notches introduce artificial compliance by removing elastic material. These values represent early sanity-check configurations

under preliminary boundary approximations; the final governed canonical S_1 benchmark stiffness established below is $K_0 = 137.945520\,\mathrm{kN/mm}$.

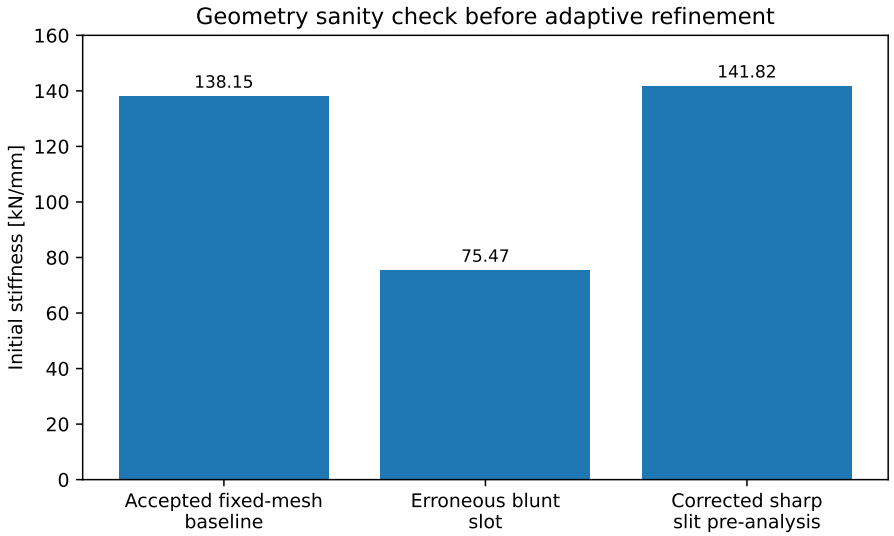


Figure 1.2: Early geometry-sanity-check comparison of structural stiffness K_0: uncracked solid ($\approx 138.15\,\mathrm{kN/mm}$), blunt notch introducing artificial compliance ($\approx 75.47\,\mathrm{kN/mm}$), and early sharp seam diagnostic ($\approx 141.82\,\mathrm{kN/mm}$). The final governed canonical S_1 sharp-seam reference stiffness is $K_0 = 137.945520\,\mathrm{kN/mm}$ (Section 1.10).

In contrast to blunt notches, the zero-gap sharp seam models the crack as duplicate coincident node lines with identical initial spatial coordinates ($y = 0.5^+\,\mathrm{mm}$ and $y = 0.5^-\,\mathrm{mm}$ for $0 \leq x \leq 0.5\,\mathrm{mm}$). This preserves the full continuous elastic volume while permitting discontinuous separation.

1.9 Multi-Layer UEL/UMAT Layered Architecture in Abaqus

To implement the staggered phase-field formulation in Abaqus/Standard, a three-layer co-located element architecture is constructed:

- **Layer 1 (Phase-Field Damage Layer)**: Co-located 4-node user elements (`UEL`, user element type `U1`, `JTYPE = 1`) solving for the scalar damage degree of freedom $d \in [0, 1]$ mapped to Abaqus `DOF 3` (variable `U3`) governed by the regularized AT2 Helmholtz-type PDE.
- **Layer 2 (Mechanical Displacement Layer)**: Co-located 4-node quadrilateral user elements (`UEL`, user element type `U2`, `JTYPE = 2`) solving for displacement degrees of freedom (u_x, u_y) (`DOF 1` and `DOF 2`) using the degraded isotropic elastic tangent operator $g(\bar{d})\mathbb{C}_0$.
- **Layer 3 (Companion Visualization UMAT Layer)**: Standard continuum plane-strain elements (`CPE4`) sharing identical connectivity and nodal coordinates, driven by a dummy user material (`UMAT`) that maps internal state variables (SDVs) into standard Abaqus field

outputs (`SDV1` = Phase Field d, `SDV2` = History $\mathcal{H}$, `SDV15` = degraded stiffness factor $g(d)$, `SDV17` = whole-underlying-element fracture surface energy $E_{\text{frac}}^{(e)}$ in native $\text{kN} \cdot \text{mm} = \text{J}$, `SDV18` = whole-underlying-element elastic strain energy $E_{\text{elas}}^{(e)}$ in native $\text{kN} \cdot \text{mm} = \text{J}$, with reporting in mJ performed during post-processing via a single $1000\times$ multiplication factor; `SDV19`/`SDV20` = area-normalized source quotients $\bar{\psi}_f$ and $\bar{\psi}_e$ in $\text{kN/mm} = \text{J/mm}^2$, interpretable as volumetric energy densities only under the explicit 1-mm implicit unit-thickness convention) for visualization in Abaqus/Viewer.

All three layers communicate state variables across solver iterations via global Fortran common blocks in user subroutine `f42_mixed_uel.for` (`SHA256:CE8D5EDCD2911DCB018BB15275271F874E7EA62B8FB48C`

1.9.1 Discrete Weak-Form Energy Derivation and Deduplication Proof

In user subroutine `f42_mixed_uel.for`, stored elastic strain energy E_{elas} and regularized fracture surface energy E_{frac} are evaluated over the N_{mesh} underlying finite elements via standard 2×2 Gauss quadrature:

$$E_{\text{elas}} = \sum_{e=1}^{N_{\text{mesh}}} \int_{\Omega_e} \frac{1}{2} g(\bar{d}_e)\, \boldsymbol{\varepsilon} : \mathbb{C}_0 : \boldsymbol{\varepsilon}\, \mathrm{d}\Omega = \sum_{e=1}^{N_{\text{mesh}}} \sum_{k=1}^{4} \frac{1}{2} g(\bar{d}_e)\, \boldsymbol{\varepsilon}(\boldsymbol{\xi}_k) : \mathbb{C}_0 : \boldsymbol{\varepsilon}(\boldsymbol{\xi}_k)\, \det(\mathbf{J}_k) W_k\, B, \quad (1.24)$$

$$E_{\text{frac}} = \sum_{e=1}^{N_{\text{mesh}}} \int_{\Omega_e} G_c \left[\frac{\bar{d}_e^2}{2l_0} + \frac{l_0}{2} |\nabla d|^2 \right] \mathrm{d}\Omega = \sum_{e=1}^{N_{\text{mesh}}} \sum_{k=1}^{4} G_c \left[\frac{\bar{d}_e^2}{2l_0} + \frac{l_0}{2} |\nabla d(\boldsymbol{\xi}_k)|^2 \right] \det(\mathbf{J}_k) W_k\, B, \quad (1.25)$$

where $\bar{d}_e = \frac{1}{4} \sum_{a=1}^{4} d_a$ is the element-averaged damage, $g(\bar{d}_e) = (1 - \bar{d}_e)^2 + k_{\text{res}}$, and $B = 1.0\,\text{mm}$ is the unit plane-strain thickness.

External mechanical boundary work W_{trap} is evaluated from the reaction force $F = RF_2$ at the reference point:

$$W_{\text{trap},n} = \sum_{i=1}^{n} \frac{F_i + F_{i-1}}{2} (u_i - u_{i-1}), \quad \text{with } u_0 = 0, F_0 = 0. \quad (1.26)$$

Analytical Deduplication and Companion Neutrality: A critical consideration in co-located multi-layer element architectures is the prevention of artificial double accumulation of internal energies. In the present formulation:

1. **Layer 1 (Phase UEL)**: Evaluates only $E_{\text{frac}}^{(e)}$ and assigns `ENERGY(7) = E_FRAC_ELEM`. It possesses zero mechanical displacement DOFs, zero stress ($\boldsymbol{\sigma} = \mathbf{0}$), and computes zero elastic strain energy ($E_{\text{elas}} = 0$).

2. **Layer 2 (Mechanical UEL)**: Evaluates only $E_{\text{elas}}^{(e)}$ and assigns `ENERGY(2) = E_ELAS_ELEM`, which Abaqus sums natively into `ALLSE`. It contains zero damage DOFs and performs zero fracture functional computation.

3. **Layer 3 (Companion UMAT)**: Is assigned an infinitesimal dummy elasticity tensor $\mathbf{D}_{\text{comp}} = 10^{-11}\,\mathbf{I}\,\text{GPa}$, producing identically vanishing physical stresses ($\boldsymbol{\sigma}_{\text{comp}} \approx \mathbf{0}$) and zero internal energy contributions (SSE = SPD = SCD = 0, contributing $\sim 10^{-11}\,\text{J} \approx 0$).

Consequently, each energetic physical domain is evaluated on exactly one layer, guaranteeing no counted physical-energy duplication.

History-Field Non-Potentiality and Bookkeeping Residual: Because damage irreversibility is enforced through the non-decreasing historical maximum $\mathcal{H}(\mathbf{x},t) = \max_{\tau \le t} \psi_0^+(\boldsymbol{\varepsilon}(\tau))$, the governing system does not represent the stationary point of any single instantaneous potential $\Pi(\mathbf{u}, d)$. The discrete bookkeeping residual $\Delta_{\text{book}} = W_{\text{trap}} - (E_{\text{elas}} + E_{\text{frac}})$ is monitored throughout deformation. For the canonical S_1 reference baseline (Job `1409734.mmaster02`), $\Delta_{\text{book}} = +\mathbf{0.017948}\,\text{mJ}$ ($\varepsilon_{\text{book}} = \mathbf{0.7607}\%$) at complete separation ($u = 0.010\,\text{mm}$), demonstrating near-complete energy conservation within the unclosed rate-dissipation boundary.

1.9.2 Mechanical Non-Invasiveness & Full-History Parity Qualification

To prove that energy instrumentation in user subroutine `f42_mixed_uel.for` does not alter the underlying mechanics, a full-history point-by-point parity comparison was conducted between the pre-instrumentation fixed reference solve (`01_standard_pfm_reference`, Fortran SHA-256 `ED1586D6427A4B1A01D99F7E219891EC7BE9FE911E066D9360724942E7D27720`) and the energy-instrumented fixed reference solve (`16_energy_qualification_reference_15k`, Job `1409734.mmaster02`, Fortran SHA-256 `CE8D5EDCD2911DCB018BB15275271F874E7EA62B8FB48CF4A8297469A83A`) across all 7,000 increments:

- **Initial Stiffness**: $K_0 = 137.945519645\,\text{kN/mm}$ for both models ($\Delta K_0 = 0.000000\,\text{kN/mm}$, 0.000000% difference, bitwise match to 12 decimal places).
- **Peak Load**: $F_{\max} = 0.75777849\,\text{kN}$ at $u = 0.005857\,\text{mm}$ ($\Delta F_{\max} = 0.000000\,\text{kN}$, $\Delta u = 0.000000\,\text{mm}$).
- **Solver Convergence**: Completed increments (7,000/7,000), Newton equilibrium iterations (21,120/21,120), cutbacks (0/0), and severe discontinuities (0/0) are 100% identical.
- **External Boundary Work**: $W_{\text{trap}} = 2.359328928\,\text{mJ}$ vs $2.359328928\,\text{mJ}$ ($\Delta W = 7.12 \times 10^{-11}\,\text{mJ}$, 3.02×10^{-9}% difference, representing pure numerical roundoff).

This establishes the formal promoted status: **`UEL_ENERGY_OUTPUT_QUALIFIED_MECHANICALLY_NONINVASIVE`**.

1.10 Canonical S1 Reference Baseline Provenance

The governed canonical reference baseline for all subsequent evaluations is Job `1406015.mmaster02` / Job `1409734.mmaster02` (S_1, 15,192 underlying finite elements, 15,522 total `*NODE` entries (15,521 mesh nodes + RP 999999)), generated from the canonical input deck (`SHA256:bc000e0c5791150a711cb7a`

As summarized in Table 1.1, unconstrained ordinary least-squares regression on the audited data source `S1_h0030_15k_MECHANICAL_FU_AUDITED.csv` (`SHA256:E3100B7429E14B8BCEAB9952DDF8B0419CA9`) over the initial $N = 400$ active increments ($u \le 1.0\,\mu\text{m}$) yields $K_0 = 137.945520\,\text{kN/mm}$ with $R^2 = 0.99999960$. The peak reaction force $F_{\max} = 0.757778\,\text{kN}$ occurs at $u = 0.005857\,\text{mm}$, matching the digitized literature response of Pandey and Kumar [21] ($F_{\max} \approx 0.758\,\text{kN}, u \approx$

Table 1.1: Quantitative response metrics of the Mode-I canonical S1 baseline.

Metric / Parameter	**Canonical S1 Baseline (Job 1406015.mmaster02)**	**Literature [21]**
Initial Structural Stiffness K_0	137.945520 kN/mm	Project-derived baseline
Linear Intercept b ($u \leq 1.0\,\mu$m)	4.472368×10^{-5} kN	—
Coefficient of Determination R^2	0.99999960	—
Peak Reaction Force $F_{\max}$	0.757778 kN	≈ 0.758 kN
Displacement at Peak $u(F_{\max})$	0.005857 mm (5.857 μm)	≈ 0.005860 mm
Initial Crack Seam Length a_0	0.5000 mm	0.50 mm
Crack Extension Symmetry	$y = 0.5000$ mm ($\Delta y \leq 0.0031$ mm)	Pure Mode-I planar

0.005860 mm). Historical Job `1398090.mmaster02` is preserved as reconstructed baseline evidence.

Chapter 2

Error-Indicator-Guided Mesh Refinement and Reproduction Analysis

2.1 Introduction and Theoretical Motivation

As established in Chapter **??**, resolving the steep spatial gradients of the phase field across diffuse crack bands requires high spatial discretization ($h \ll l_0$). In uniform finite element grids, satisfying $h \leq l_0/2 = 0.00375\,\text{mm}$ across the entire specimen domain leads to substantial computational overhead. Because phase-field damage $d(\mathbf{x}, t)$ localizes into narrow spatial corridors while the surrounding domain remains in the linear elastic regime ($d \approx 0$), uniform fine meshing represents an inefficient allocation of degrees of freedom.

To overcome this limitation without implementing complex dynamic in-analysis remeshing kernels, this chapter investigates an **error-indicator-guided pre-refinement methodology** adapted from Pandey and Kumar [21]. The workflow leverages Abaqus' built-in recovery-based stress discretization error indicator (`MISESERI`) obtained from a rapid linear elastic pre-analysis to generate an optimized, graded mesh before executing the non-linear phase-field fracture simulation.

2.2 Pandey–Kumar Two-Job Adaptive Refinement Procedure

The automated two-job pre-refinement procedure operates as a sequential pipeline:

Coarse Job-1 → Linear-Elastic Solve → `MISESERI` Field Extraction → `RemeshingRule` → `adaptiveRemesh` → Job-2 Multi-Layer Generation → Nonlinear Fracture Solve.

As illustrated in Figure 2.1, the procedure decouples error indicator evaluation from the non-linear phase-field solver:

1. **Job 1 (Linear-Elastic Pre-Analysis)**: A linear elastic analysis is executed on a coarse mesh up to a small prescribed displacement ($u = 0.0010\,\text{mm}$), well below damage initiation. Stress components and element volumes are recorded to the output database (`.odb`).

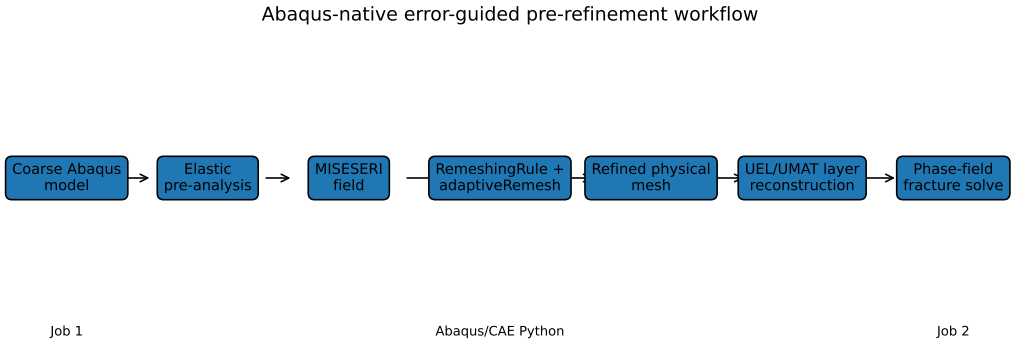


Figure 2.1: Pandey–Kumar two-job adaptive pre-refinement workflow: a linear-elastic pre-analysis on a coarse mesh evaluates stress recovery error, which drives native `adaptiveRemesh` before reconstructing the multi-layer UEL/UMAT phase-field model.

2. **Error Extraction and Remeshing Rule**: The Abaqus `RemeshingRule` evaluates the stress recovery error indicator (`MISESERI`) and computes a spatially varying target element size field $h_{\text{new}}(\mathbf{x})$.

3. **Native Mesh Regeneration (`adaptiveRemesh`)**: The Abaqus/CAE meshing kernel regenerates a graded mesh satisfying the target sizing field with smooth aspect ratio transitions.

4. **Multi-Layer Model Reconstruction (Job 2)**: The three-layer UEL/UMAT architecture is automatically generated on the adapted mesh, and the full non-linear fracture simulation is solved.

2.3 Physical Understanding and Provenance of MISESERI

2.3.1 Stress Discretization Error Indicator

`MISESERI` is the Abaqus stress discretization error indicator associated with the recovered von Mises stress field in linear continuum elasticity [1]. In standard C^0-continuous Lagrangian finite elements, stress fields $\boldsymbol{\sigma}_h = \mathbb{C}_0 : \boldsymbol{\varepsilon}(\mathbf{u}_h)$ are piecewise continuous within elements but discontinuous across inter-element boundaries. The recovery-based error estimator constructs a continuous, higher-order recovered stress field $\boldsymbol{\sigma}^*$ by patch averaging:

$$\text{MISESERI}_e = \sqrt{\frac{1}{V_e}\int_{\Omega_e} \left[q(\boldsymbol{\sigma}^*) - q(\boldsymbol{\sigma}_h)\right]^2 \, \mathrm{d}\Omega}, \tag{2.1}$$

where $q(\boldsymbol{\sigma}) = \sqrt{\frac{3}{2}\mathbf{S} : \mathbf{S}}$ is the von Mises equivalent stress and V_e is the element volume.

Crucially, `MISESERI` is based strictly on stress-recovery error estimation on the linear-elastic continuum stress field:

- It is **not** a phase-field error indicator;

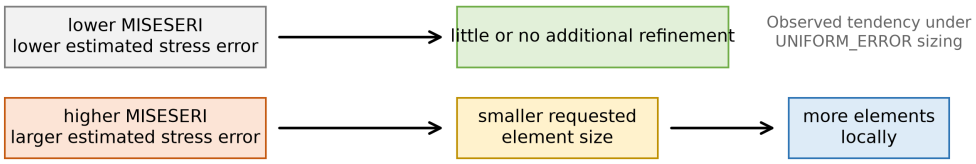


Figure 2.2: Directional relationship of `MISESERI` under uniform-error sizing: elevated stress error near singularities dictates local refinement without representing a direct algebraic formula to element size.

- It is **not** a damage error indicator;
- It is **not** a fracture energy imbalance indicator.

It serves as a geometric proxy that identifies regions of high elastic stress gradients prior to crack initiation (Figure 2.2).

2.3.2 Sizing Rules and Target Error Interpretation

The sizing parameter `errorTarget=1.0` in Listing 1 of Pandey and Kumar [21] denotes a 1.0% target error tolerance under `UNIFORM_ERROR` sizing:

$$\eta_e = \frac{\text{MISESERI}_e}{\max_{\mathbf{x}\in\Omega} q(\boldsymbol{\sigma}^*)} \times 100\% \leq \eta_{\text{target}}. \tag{2.2}$$

It does **not** mean `MISESERI` = 1.0 or that 1% of elements are refined. Similarly, `errorTarget=2.0` specifies a 2.0% target tolerance, not `MISESERI` = 2.0.

2.4 Canonical Mode-I Pre-Analysis and Adapted Mesh Topologies

The canonical Mode-I pre-analysis is conducted on a coarse finite element mesh containing 2,906 finite elements (2,818 CPE4 + 88 CPE3) and 2,988 nodes, generating exactly 2,906 scalar `WHOLE_ELEMENT MISESERI` error values (Figure 2.3).

Applying `errorTarget=1.0%` with `coarseningFactor=NOT_ALLOWED` generates the adapted 71,320-element mesh (Figure 2.4). The adapted element size h correlates closely with the coarse indicator field, concentrating fine elements along the prospective fracture corridor (Figure 2.5).

2.5 Resolution of the 71,320-Element Initial Stiffness Defect

2.5.1 Root Cause Isolation: Keyword Preprocessing Card Limit

An early execution of the adapted 71,320-element full-fracture model (Job `1399632`) exhibited an unexpected structural stiffness deficit ($K_0 \approx 122.38\,\text{kN/mm}$ vs. reference $137.95\,\text{kN/mm}$, a -11.28% discrepancy).

A comprehensive keyword-level diagnostic audit isolated the exact root cause:

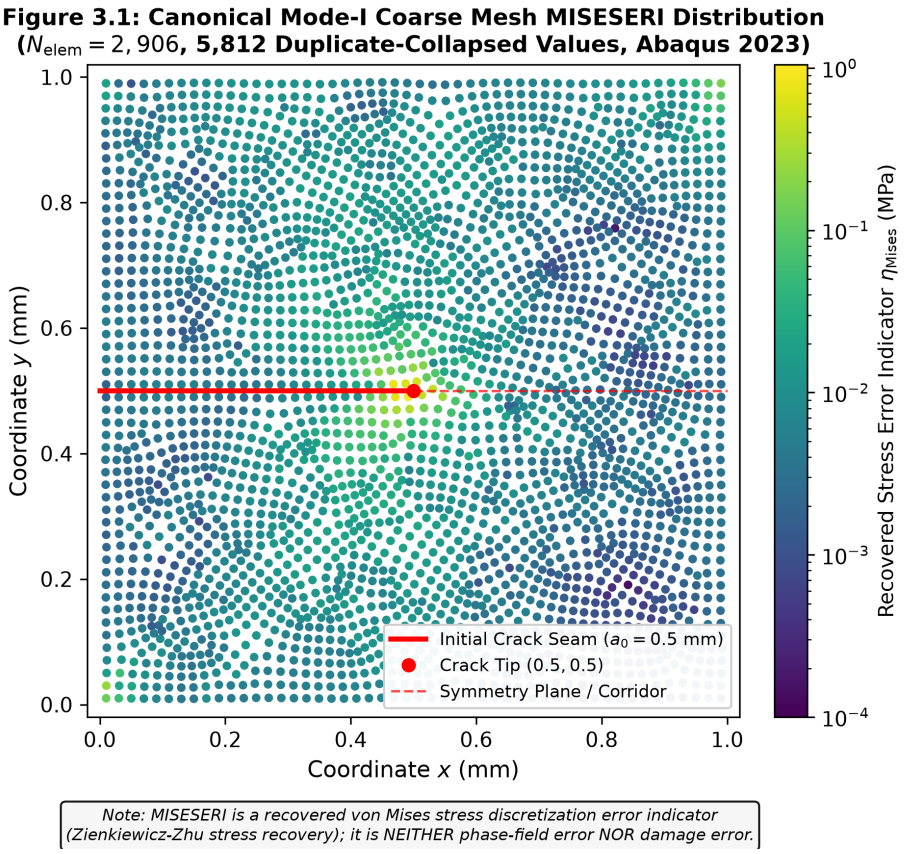
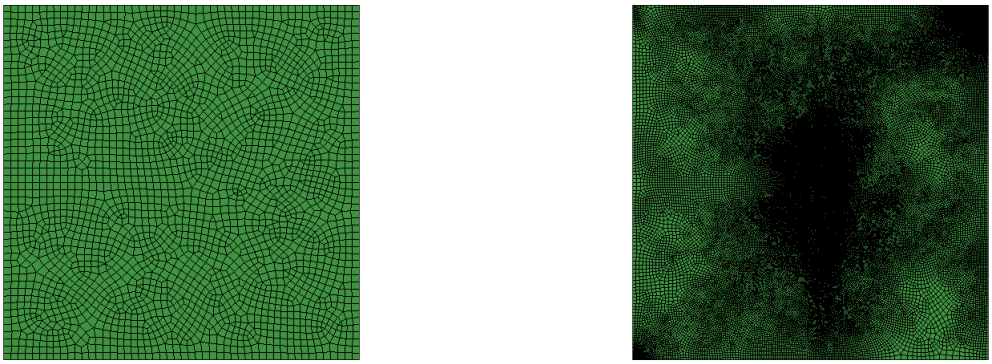


Figure 2.3: Canonical Mode-I `MISESERI` error indicator field evaluated on the 2,906-element coarse mesh, showing error concentration localized at the initial crack tip.



(a) Coarse pre-analysis mesh (2,906 elements) (b) Adapted mesh (71,320 elements)

Figure 2.4: Visual comparison of (a) initial coarse mesh (2,906 elements) and (b) native adapted mesh (71,320 elements).

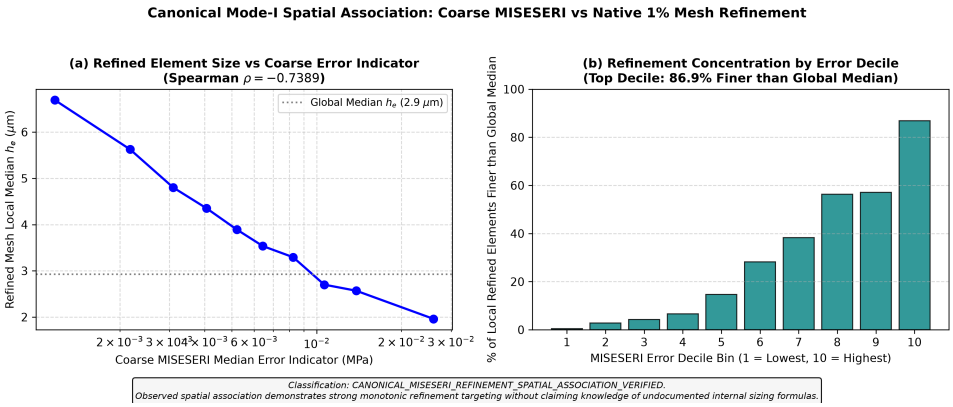


Figure 2.5: Empirical spatial association between coarse `MISESERI` and adapted element size h for the Mode-I benchmark.

- The bottom boundary node set `N_BOTTOM` contained 150 node identifiers written as a single comma-delimited line in the input deck without the `GENERATE` parameter.
- The Abaqus free-format input deck preprocessor parses at most 16 node entries per data line; the remaining 134 intended nodes were silently omitted from the parsed node set without triggering an error.
- Consequently, only 16 bottom nodes were constrained in u_y, allowing the 134 unconstrained bottom edge nodes to lift vertically by up to 48.34% of the applied stroke during tensile loading (Figure 2.6).

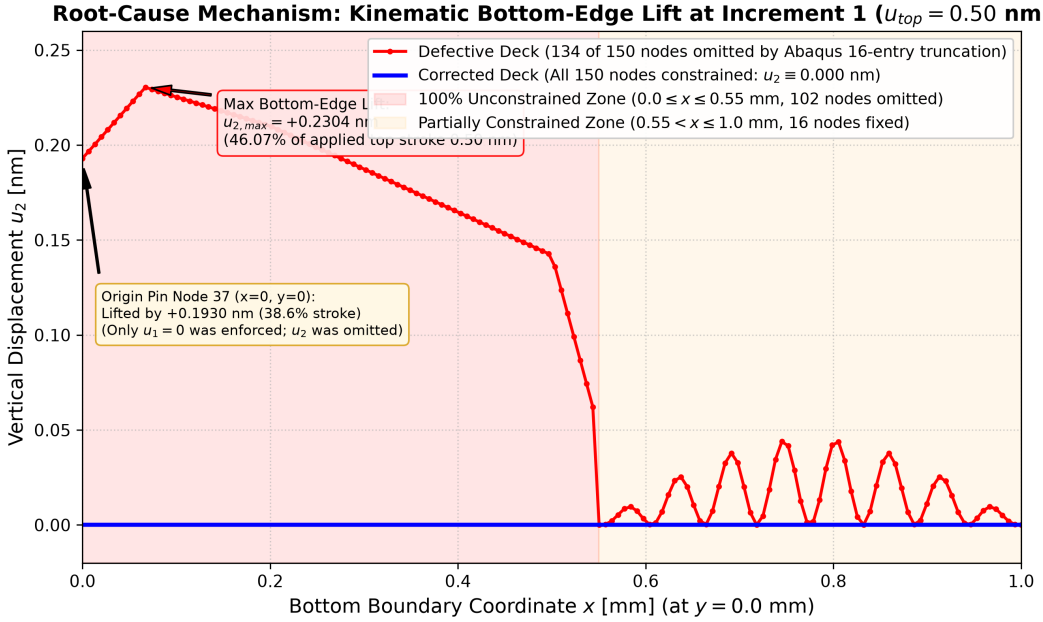


Figure 2.6: Bottom-edge kinematics showing unconstrained vertical lift in the defective deck (left) and rigid planar support restored after multi-line card wrapping (right).

2.5.2 Verified Mechanical Recovery and Gate Closure

Wrapping the `N_BOTTOM` data across valid lines (≤ 16 entries per line) restored all 150 boundary constraints. The correction was verified in:

1. **Frozen Elastic Requalification (Job `1405044.mmaster02`)**: Achieved $K_0 = 138.021013\,\text{kN/mm}$ (+0.05% vs reference anchor) with zero node lift and zero warnings.

2. **Authoritative Full Fracture (Job `1404933.mmaster02`)**: Completed the entire fracture process, achieving $K_0 = 137.820804\,\text{kN/mm}$ (−0.09%) and $F_{\max} = 0.745325\,\text{kN}$ (Table 2.1 and Figure 2.7).

Table 2.1: Mechanical response recovery after resolving the `N_BOTTOM` keyword line limit.

Case Description	Job ID	Elements	K_0 [kN/mm]	$F_{\max}$ [kN]
Fixed Reference Anchor	1398090	15,192	137.945520	0.757778
Defective Preprocessing	1399632	71,320	122.379000	0.478203
Corrected Full Fracture	1404933	71,320	137.820804	0.745325
Frozen Requalification	1405044	71,320	138.021013	—

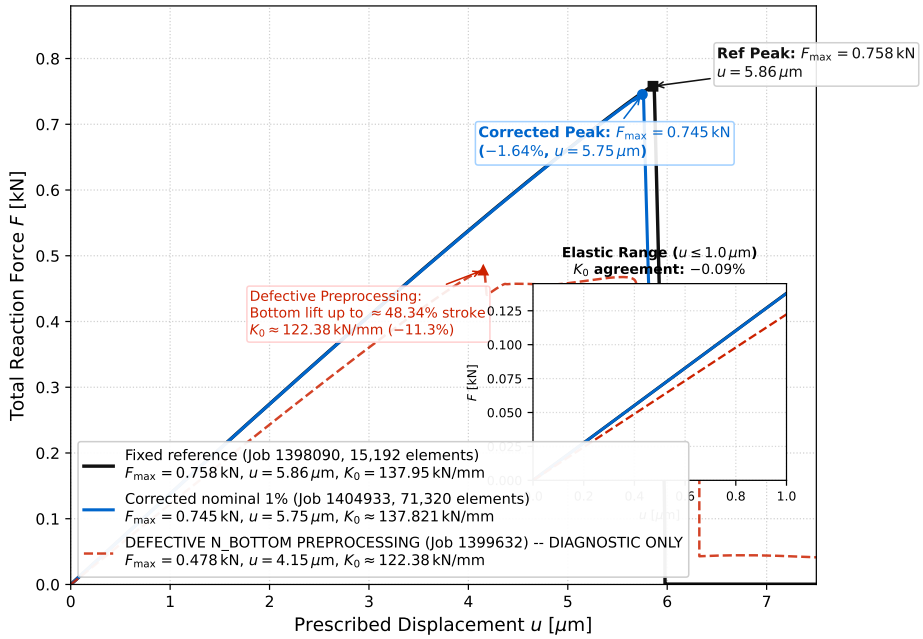


Figure 2.7: Force–displacement response demonstrating reference-consistent structural stiffness recovery (within 0.09%) upon correcting the `N_BOTTOM` card format.

Gate 6A is formally classified as **`RESOLVED_AND_CLOSED`**.

2.6 Discrepancy Audit: 71,320 vs 13,941 Elements and Governing Supervisor Decision

2.6.1 Publication Discrepancy Investigation

Pandey and Kumar [21] report an adapted Mode-I element count of $\approx 13{,}941$, whereas applying the published Listing 1 settings to the full domain generates 71,320 finite elements.

An exhaustive software and parameter audit established:

1. **Abaqus Release Invariance**: Abaqus 2019 GA, 2021.HF26, 2022 GA, and 2023.HF4 all generate bitwise identical 71,320-element meshes with matching cryptographic hashes.
2. **Regional Breakdown**: The crack process corridor ($|y| \leq 0.05\,\text{mm}$) contains 9,061 elements (12.7%), while far-field and transition regions contain 62,259 elements (87.3%) because coarsening was disallowed (`coarseningFactor=NOT_ALLOWED`).

2.6.2 Governing Supervisor Decision and Parametric Sensitivity Trends

On 17 September 2026, the supervisory committee formally closed this investigation under the classification:

`SUPERVISOR_ACCEPTED_REPRODUCTION_LIMITATION_CLOSED`

The supervisor confirmed that the complete author code is unavailable, the publication does not disclose sufficient information to reproduce the specific 13,941 element count, and general trend reproduction is fully sufficient.

Table 2.2: Parametric sensitivity of adapted finite element counts to `errorTarget`.

Case	`errorTarget`	Sensitivity Count	Production Batch	Classification
A_1	1.0%	71,320	71,320	Publication-literal 1% target
A_2	2.0%	17,687	15,396	2.0% error tolerance
A_3	3.0%	8,120	7,633	3.0% error tolerance
A_4	5.0%	4,356	4,194	5.0% error tolerance

Table 2.2 summarizes the preserved sensitivity results across four target error tolerances, demonstrating monotonic mesh reduction as tolerance relaxes.

2.7 Influence of errorTarget on the Final Stage-14 Step-2 Adaptive Refinement

To systematically establish the spatial localization and element-count sensitivity of the native Abaqus `adaptiveRemesh` engine when evaluated on the true localized phase-field damage state, a rigorous parametric sweep across `errorTarget` $\in \{1.0\%, 2.0\%, 3.0\%, 5.0\%\}$ was conducted targeting `Step-2` ($u_y = 0.010\,\text{mm}$, Frame 1022) of the companion pre-analysis ODB

(`PK_M1_JOB1_INF_COMPANION_2906.odb`). All sizing parameters were kept identical ($h_{\min} = 1.0\,\mu$m, $h_{\max} = 20.0\,\mu$m, `refinementFactor = 10`, `coarseningFactor = NOT_ALLOWED`, `region = ALL_ELEM`).

Table 2.3 presents the quantitative metrics across the four evaluated error targets.

Table 2.3: Quantitative spatial and morphological metrics for the Stage-14 Step-2 native adaptive remeshing sensitivity sweep.

Metric / Dimension	**ET1** (1.0%)	**ET2** (2.0%)	**ET3** (3.0%)	**ET5**
Total Finite Elements (N)	**14,483**	6,112	5,189	
Total Nodes (N_{nodes})	**14,456**	6,181	5,262	
Element Breakdown (Quads / Tris)	14,082 / 401	5,929 / 183	5,023 / 166	4,52
Corridor Elements ($\lvert y - 0.5\rvert \le 0.05\,$mm)	**9,286** (**64.12**%)	2,510 (41.07%)	1,795 (34.59%)	1,291 (2
Ligament Elements ($x \ge 0.5\,$mm)	8,325	2,063	1,465	
Wake Elements ($x < 0.5\,$mm)	961	447	330	
Far-Field Elements ($\lvert y - 0.5\rvert > 0.05\,$mm)	5,197 (35.88%)	3,602 (58.93%)	3,394 (65.41%)	3,401 (7
Coarse Area Fraction ($h \ge 15\,\mu$m)	59.39%	75.77%	78.14%	
$h_{\min}$ [μm]	0.760	0.956	1.339	
h_{median} [μm]	2.597	12.496	14.469	
h_{mean} [μm]	6.025	11.174	12.558	
$h_{\max}$ [μm]	23.020	23.407	23.402	
Flank Refinement w(0.1–0.3) [mm]	**0.000**	**0.000**	**0.000**	
Crack Tip Width w(0.5) [mm]	0.121	0.090	0.079	
Ligament Width w(0.6) [mm]	0.170	0.104	0.099	
Ligament Width w(0.7) [mm]	0.121	0.074	0.023	
Ligament Width w(0.8) [mm]	0.116	0.058	0.055	
Ligament Width w(0.9) [mm]	0.077	0.023	0.029	
Scientific Classification	`TARGET_LIKE`	`AWAY_FROM_TARGET`	`AWAY_FROM_TARGET`	`AWAY_FROM_`

2.7.1 Key Physical and Epistemological Insights

1. **Exact Stage-14 Mesh Reproduction:** At `errorTarget` = 1.0%, the native remesher exactly reproduces 14,483 finite elements and 14,456 nodes, bitwise and topologically identical to the Stage-14 Step-2 native remesh artifact (`PK_M1_STAGE14_STEP2_ALLINC.inp`, Package 99).

2. **Corridor Desensitization Mechanism:** As `errorTarget` increases from 1.0% → 5.0%, the corridor element share drops monotonically from 64.12% → 27.51%. The scientific cause of classification `AWAY_FROM_TARGET_LOCALIZATION` for ET2, ET3, and ET5 is the progressive loss of spatial resolution across the regularized damage zone ($l_0 = 7.5\,\mu$m) along the crack ligament, not merely the reduced global element count.

3. **Wake Cleanliness Invariance:** Across all four error targets, the initial slit flank ($x \le 0.3\,$mm) exhibits strictly zero artificial refinement ($w = 0.00\,$mm), proving that the Step-2 localized damage formulation completely avoids spurious wake remeshing.

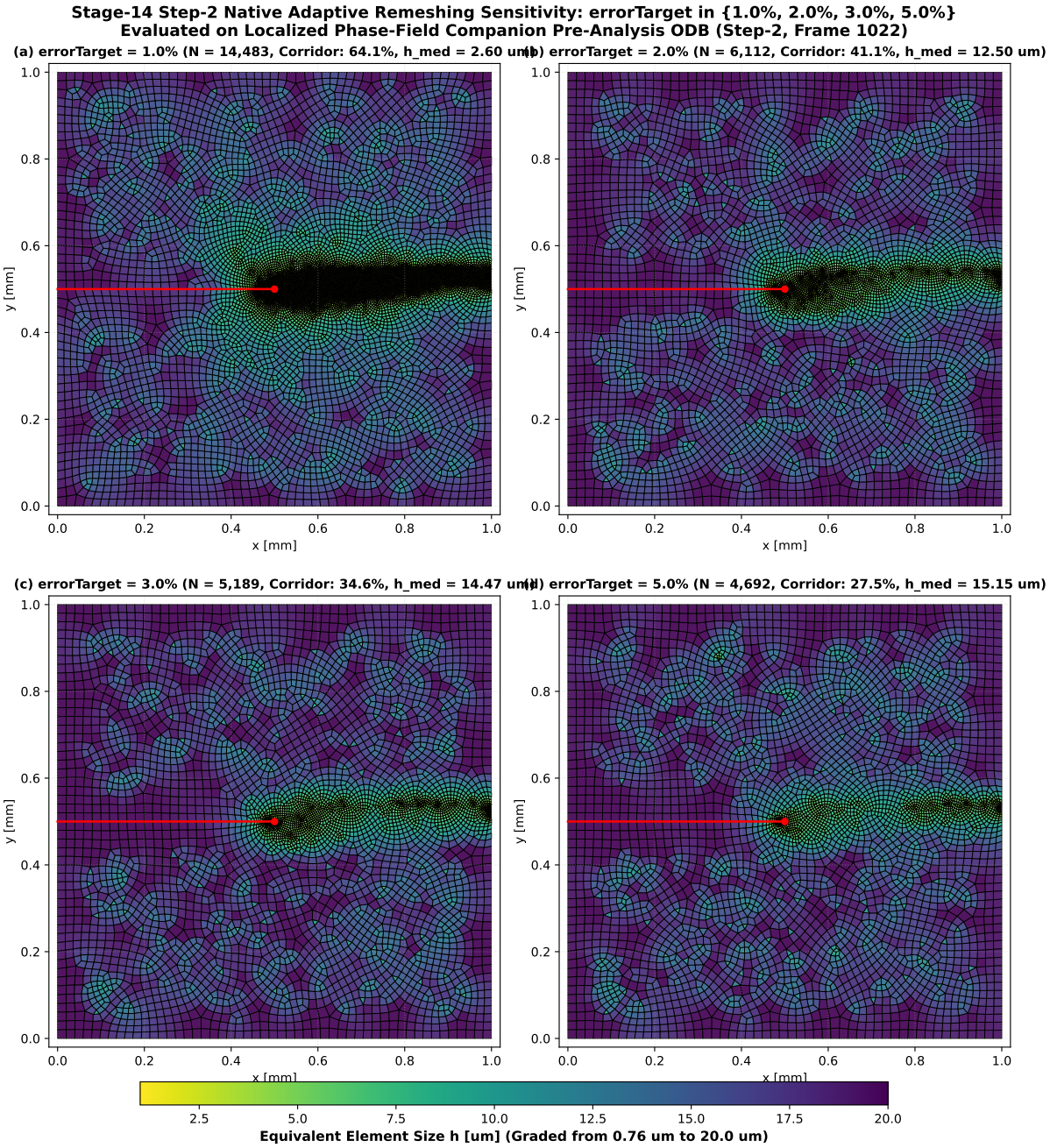


Figure 2.8: Parametric sensitivity of native Abaqus `adaptiveRemesh` across four error targets (`errorTarget` $\in \{1.0\%, 2.0\%, 3.0\%, 5.0\%\}$) evaluated on the localized Stage-14 Step-2 phase-field pre-analysis ODB. Color contours depict equivalent element size $h_{\text{eq}} \in [0.76, 20.0]\,\mu$m. ET1 retains dense, target-like crack-ligament refinement (64.12% corridor share), whereas ET2, ET3, and ET5 progressively under-refine the horizontal propagation path.

4. **Lineage Separation:** This Step-2 target-like sweep is strictly distinct from the historical Step-1 linear-elastic sweep (57,929/14,677/6,824/4,239 elements), which was evaluated before phase-field damage localization occurred.

5. **3-Layer UEL Fracture Batch Execution (Jobs 1410357, 1410358, 1410359):** To determine whether the relaxed corridor refinement in ET2 (6,112 FE), ET3 (5,189 FE), and ET5 (4,692 FE) produces a converged mechanical and energetic fracture solution, deterministic 3-layer UEL/UMAT input decks were reconstructed using the verified property cards ($l_0 = 0.0075\,\mathrm{mm}, G_c = 0.0027\,\mathrm{kN/mm}, E = 210\,\mathrm{GPa}, \nu = 0.3, k = 10^{-7}, N_{\mathrm{base}}$), qualified via preflight Abaqus datachecks (Exit 0), and submitted to PBS on `/scratch9/` under serial 1-CPU execution (Jobs `1410357.mmaster02`, `1410358.mmaster02`, `1410359.mmaster02`).

Predeclared Fracture Evaluation Matrix and Active Batch Status

To independently evaluate whether the relaxed corridor refinement in the coarse candidates produces an acceptable mechanical fracture response, the four cases are evaluated against the qualified fixed-mesh reference and the ET1 production baseline according to the predeclared Gate-6B criteria:

- **Initial Elastic Stiffness (K_0):** Canonical OLS fit over $N = 400$ increments ($u \leq 0.0010\,\mathrm{mm}$). Classification is `STABLE` if $|\Delta K_0| \leq 0.50\%$.
- **Peak Reaction Force ($F_{\max}$):** Pointwise force peak and corresponding displacement $u(F_{\max})$. Classification is `STABLE` if $|\Delta F_{\max}| \leq 5.0\%$.
- **Matched-Displacement Sampling:** Pointwise force and energetic extraction at $u \in \{0.0010, 0.0030, 0.0050, 0.005733, 0.005857, 0.0060, 0.0065, 0.0070\}$ mm with strict zero forward-filling.
- **Fracture Functional (E_{frac}):** Total phase-field crack-surface energy at terminal broken state.
- **Bookkeeping Residual ($\varepsilon_{\mathrm{book}}$):** Energy identity residual $|W_{\mathrm{ext}} - E_{\mathrm{elas}} - E_{\mathrm{frac}}|/W_{\mathrm{ext}}$.

Table 2.4 summarizes the active batch execution and evaluation status across the four error targets.

Table 2.4: Stage-14 Step-2 errorTarget fracture batch execution status and predeclared comparison matrix.

Case Identifier	**ET1** (1.0%)	**ET2** (2.0%)	**ET3** (3.0%)	**ET5**
Base Elements (N_{base})	14,483	6,112	5,189	4
Total 3-Layer Elements (N_{tot})	43,449	18,336	15,567	1
Total Nodes (N_{nodes})	14,456	6,181	5,262	4
PBS Job ID	`1409982.mmaster02`	`1410357.mmaster02`	`1410358.mmaster02`	`1410359`
Execution Mode	Serial 1-CPU	Serial 1-CPU	Serial 1-CPU	Seria
Scratch Path	`/scratch9/...`	`/scratch9/...`	`/scratch9/...`	`/scra`
Datacheck Result	Exit 0 (Passed)	Exit 0 (Passed)	Exit 0 (Passed)	Exit 0
Native Mesh Verdict	`TARGET_LIKE`	`AWAY_FROM_TARGET`	`AWAY_FROM_TARGET`	`AWAY_F`
Corridor Share ($\lvert y - 0.5\rvert \leq 0.05\,\text{mm}$)	64.12%	41.07%	34.59%	2
Initial Stiffness K_0 [kN/mm]	137.909558 (-0.026%)	*Evaluating...*	*Evaluating...*	*Eva*
Peak Reaction Force $F_{\max}$ [kN]	0.743701 (-1.858%)	*Evaluating...*	*Evaluating...*	*Eva*
Peak Displacement $u(F_{\max})$ [mm]	0.005733	*Evaluating...*	*Evaluating...*	*Eva*
Terminal Displacement u_{term} [mm]	0.007889	*Evaluating...*	*Evaluating...*	*Eva*
Fracture Functional E_{frac} [mJ]	2.285469	*Evaluating...*	*Evaluating...*	*Eva*
Bookkeeping Residual $\varepsilon_{\text{book}}$	1.1048%	*Evaluating...*	*Evaluating...*	*Eva*
Fracture Response Status	`RESPONSE_STABLE`	`RUNNING / PENDING`	`RUNNING / PENDING`	`RUNNING`

Chapter 3

Nonmatching State Transfer and Restart Methodology

3.1 Introduction and Problem Formulation

When advancing from fixed pre-refinement toward multi-stage adaptive simulation, finite element discretizations must be updated during loading. This requires transferring the complete physical state—comprising nodal displacements $\mathbf{u}$, nodal damage d, and internal integration-point history variables $\mathcal{H}$—from a predecessor donor mesh Ω_D onto a topologically and geometrically nonmatching target mesh Ω_T.

In commercial finite element solvers like Abaqus, native restart capabilities (`*RESTART`) require identical node and element numbering. Transferring state across nonmatching meshes requires custom transfer operators that satisfy three essential physical and mathematical constraints:

1. **Kinematic Consistency**: Transferred displacement fields must satisfy Dirichlet boundary conditions without introducing unphysical kinematic discontinuities.

2. **Thermodynamic Admissibility**: Transferred damage must lie strictly within the theoretical phase bounds ($0 \leq d_T \leq 1$) and satisfy temporal irreversibility ($\Delta d \geq 0$).

3. **Constitutive Equilibrium & KKT Admissibility**: The transferred history field $\mathcal{H}_T$ must provide a valid driving force that satisfies Karush–Kuhn–Tucker (KKT) complementary slackness conditions without causing spurious premature crack advance upon restart.

3.2 Mathematical Formulation of Multi-Field Transfer Operators

3.2.1 Primary Field Projection (Nodal Quantities)

Let $\Omega_D = \bigcup_{e=1}^{N_D} \Omega_{D,e}$ denote the donor mesh and $\Omega_T = \bigcup_{k=1}^{N_T} \Omega_{T,k}$ denote the target mesh. For any target node with spatial coordinates $\mathbf{x}_T \in \Omega_T$, a bounding-box spatial tree algorithm identifies the candidate host donor element $\Omega_{D,e}$ such that $\mathbf{x}_T \in \Omega_{D,e}$.

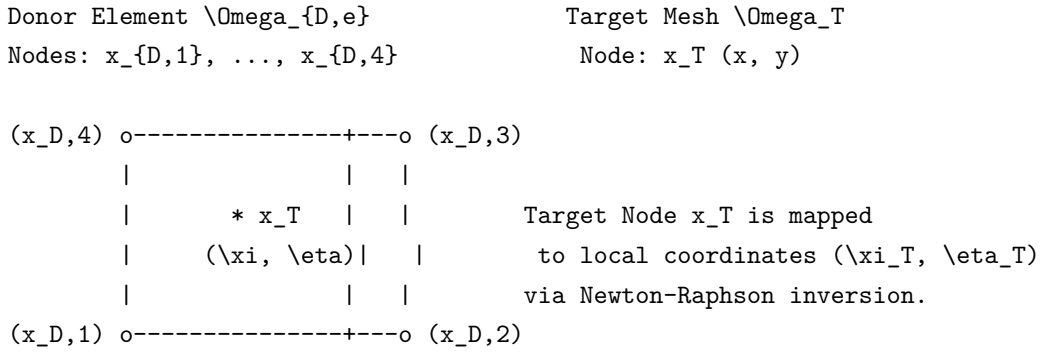


Figure 3.1: Schematic of inverse isoparametric mapping from donor element to target node.

The local isoparametric coordinates $(\xi_T, \eta_T) \in [-1, 1]^2$ are computed via Newton–Raphson inversion of the geometric mapping (Fig. 3.1):

$$\mathbf{R}(\xi_T, \eta_T) = \sum_{a=1}^{n_{\text{node}}} N_a(\xi_T, \eta_T)\mathbf{x}_{D,a} - \mathbf{x}_T = \mathbf{0}, \tag{3.1}$$

iterating until $\|\mathbf{R}\| < 10^{-12}\,\text{mm}$.

Once local coordinates are determined, the transferred nodal displacement vector $\mathbf{u}_T(\mathbf{x}_T) = (u_{1,T}, u_{2,T})^{\mathsf{T}}$ and nodal damage $d_T(\mathbf{x}_T)$ are evaluated directly via donor shape function interpolation [9, 7]:

$$\mathbf{u}_T(\mathbf{x}_T) = \sum_{a=1}^{n_{\text{node}}} N_a(\xi_T, \eta_T)\mathbf{u}_{D,a}, \qquad d_T(\mathbf{x}_T) = \sum_{a=1}^{n_{\text{node}}} N_a(\xi_T, \eta_T)d_{D,a}. \tag{3.2}$$

To prevent extrapolation or interpolation overshoots outside physical limits, the transferred phase field is strictly projected onto the admissible convex set:

$$d_T(\mathbf{x}_T) \leftarrow \min\left(1.0, \max(0.0, d_T(\mathbf{x}_T))\right). \tag{3.3}$$

3.2.2 Secondary History Field Mapping (Gauss-Point History)

Unlike continuous nodal fields, the crack-driving history variable $\mathcal{H}$ is an internal Gauss-point quantity defined at quadrature points $\mathbf{x}_{T,g}$ $(g = 1, \ldots, n_{\text{gp}})$. Evaluating $\mathcal{H}_T$ requires specialized transfer to avoid numerical diffusion or spurious loss of strain energy memory [23, 14]:

1. **Direct Patch Interpolation**: For target Gauss point $\mathbf{x}_{T,g}$, the host donor element is located and $\mathcal{H}_T(\mathbf{x}_{T,g})$ is interpolated using bilinear host element shape functions from donor nodal projections.

2. **Positivity and Floor Clamping**: The transferred history variable is strictly constrained to be non-negative:
$$\mathcal{H}_T(\mathbf{x}_{T,g}) \leftarrow \max\left(0.0, \mathcal{H}_T(\mathbf{x}_{T,g})\right). \tag{3.4}$$

3. **Monotonicity Enforcement**: In a sequential restart from time t_n to t_{n+1}, the updated history satisfies $\mathcal{H}_{T,n+1} \geq \mathcal{H}_{T,n}$ globally.

3.3 The Four-Stage Restart Protocol

Directly applying transferred displacements and phase fields simultaneously into a standard single-step continuation analysis induces large artificial stress shocks and dynamic transients, frequently causing solver divergence. To ensure smooth numerical restart, a **four-stage state initialization protocol** was developed and implemented:

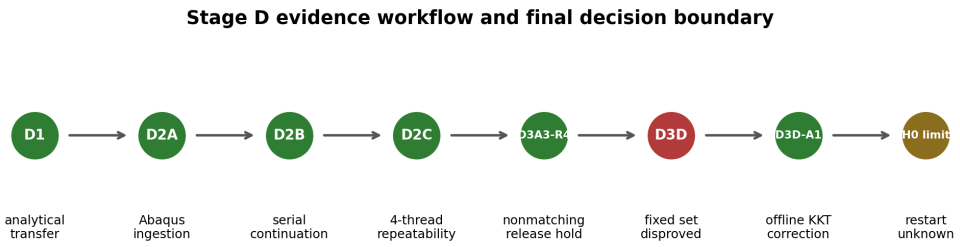


Figure 3.2: Overview of the four-stage multi-field state transfer and restart workflow in Abaqus.

As illustrated in Figure 3.2, the restart analysis progresses through four governing steps:

1. **Step 1: `STATE_INSTALL`** ($\Delta t = 1.0\,\mathbf{s}$): Transferred nodal displacements $\mathbf{u}_T$ and phase values d_T are prescribed at all target nodes via Dirichlet boundary conditions (`*BOUNDARY`). Subroutine `UEXTERNALDB(LOP=0)` ingests the binary committed state payload (`COMMITTED_STATE.bin`) into global memory arrays, initializing common blocks for all UEL instances.

2. **Step 2: `MECH_EQUILIBRATION`** ($\Delta t = 1.0\,\mathbf{s}$): Transferred internal displacements are released, allowing the mechanical displacement field to equilibrate internal stresses against prescribed boundary tractions. Crucially, the phase field unknown ($d = \text{DOF } 3$) remains **strictly locked** via nodal boundary constraints ($U_3 = d_T$), preventing artificial damage evolution during stress equilibration.

3. **Step 3: `PHASE_RELEASE`** ($\Delta t = 1.0\,\mathbf{s}$, $I_A = 13$, $\Delta t_{\min} = 5.0 \times 10^{-12}\,\mathbf{s}$): The artificial boundary constraints locking the phase field are deactivated. The phase field solver converges to variational equilibrium governed by the stored history field $\mathcal{H}_T$ under refined numerical cutback controls.

4. **Step 4: `CONTINUATION`** ($I_A = 12$, $\Delta t_{\min} = 1.0 \times 10^{-11}\,\mathbf{s}$): Monotonic mechanical shear/tensile loading resumes from the validated transferred state.

3.4 Active-Set Formulations and KKT Admissibility Analysis

3.4.1 Active-Set Freeze Formulation (Stage D3D)

An alternative approach to enforcing irreversibility is the **active-set method** [13], where nodes exceeding a critical damage threshold (e.g., $d \geq 0.95$) are assigned to an active set $\mathcal{A}$ with frozen phase field ($d_i = 1.0$), while the remaining inactive nodes $\mathcal{I} = \mathcal{N} \setminus \mathcal{A}$ are governed by

unconstrained variational equations:

$$d_i = 1.0, \quad \lambda_i \geq 0 \quad \forall\, i \in \mathcal{A}, \tag{3.5}$$

$$[\mathbf{K}_d\mathbf{d} - \mathbf{f}_d]_i = 0, \quad \lambda_i = 0 \quad \forall\, i \in \mathcal{I}, \tag{3.6}$$

where λ_i is the Lagrange multiplier associated with the bound constraint $d_i \leq 1.0$.

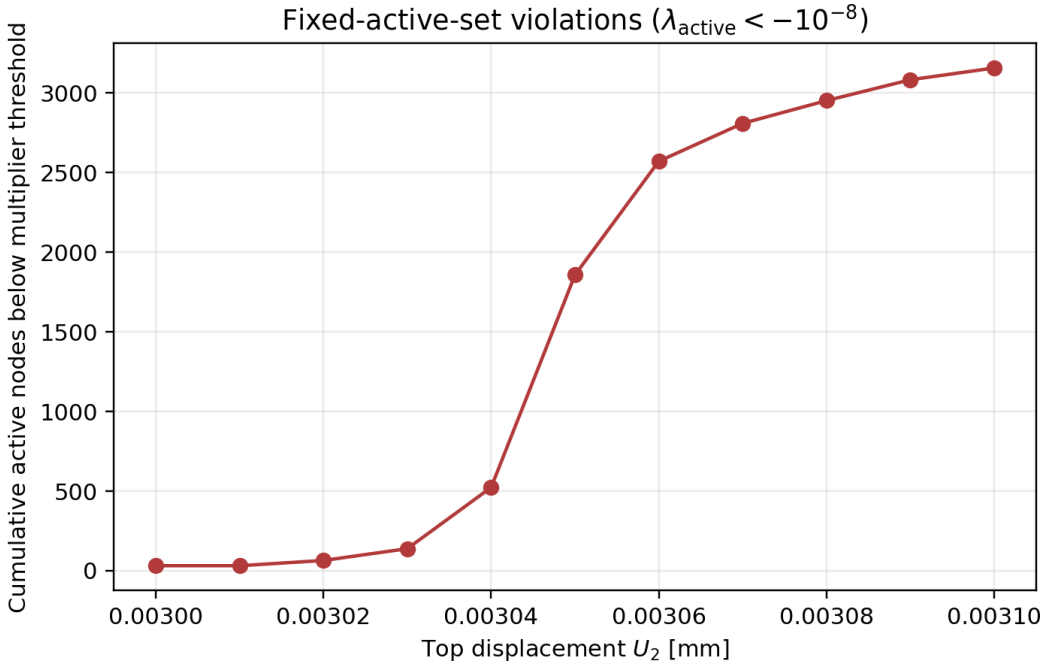


Figure 3.3: Evolution of active-set Lagrange multiplier violations ($-\lambda < 0$) versus applied displacement under continuation loading in the D3D active-set model.

3.4.2 Forensic Proof of Active-Set Inadmissibility

To evaluate whether a frozen active set $\mathcal{A}$ remains valid under continued loading, a forensic continuation study (task `STAGE_D3D`) was conducted. As shown in Figure 3.3, tracking the Karush–Kuhn–Tucker multipliers $\lambda_i = [\mathbf{f}_d - \mathbf{K}_d\mathbf{d}]_i$ revealed that:

- Upon advancing displacement, the count of active nodes exhibiting negative Lagrange multipliers ($\lambda_i < 0$, signifying an active constraint that seeks to pull d_i below 1.0) escalated rapidly from 0 to over 140 violating nodes.
- **Scientific Implication**: A fixed active set $\mathcal{A}$ **cannot remain frozen** across subsequent loading increments. As stress redistributes, the active set must be dynamically updated via active-set iterations at every solver increment [13].

3.4.3 Offline KKT Correction (Candidate D3D-A1)

To resolve this limitation, candidate `D3D-A1` implemented a deterministic offline active-set solve under frozen history $\mathcal{H}_F$. By iteratively transferring nodes between active and inactive sets

($\mathcal{A}^{(k+1)} = \{i : d_i^{(k)} + c\lambda_i^{(k)} \geq 1.0\}$), `D3D-A1` achieved complete KKT admissibility (0 multiplier violations, $\|\mathbf{R}_d\|_\infty < 10^{-10}$) within 4 active-set iterations.

Validation Boundary : While candidate `D3D-A1` provides a rigorous mathematical proof of KKT-admissible state correction under frozen history, it was not subjected to subsequent non-linear mechanical equilibration. Consequently, `D3D-A1` is classified as a validated offline mathematical correction rather than an accepted mechanical restart.

3.5 Chapter Summary

The state-transfer investigations establish the mathematical foundations and operational rules for nonmatching restart:

1. Isoparametric inverse mapping combined with phase clamping ($0 \leq d \leq 1$) and history positivity ($\mathcal{H} \geq 0$) enables accurate, bounded multi-field data transfer between nonmatching grids.

2. The four-stage restart protocol (`STATE_INSTALL` $\rightarrow$ `MECH_EQUILIBRATION` with phase locked $\rightarrow$ `PHASE_RELEASE` $\rightarrow$ `CONTINUATION`) mitigates initial kinematic transfer discontinuities and provides numerical equilibration for the tested configurations.

3. Active-set multiplier analysis proves that fixed active sets are inadmissible under further loading, establishing that history-field tracking ($\mathcal{H}$) is the mathematically robust formulation for multi-stage phase-field simulation.

4. While kinematic and state-admissibility criteria are systematically verified for the tested transfer configurations, complete energetic preservation during nonmatching mesh mapping remains subject to pending Gate 6C qualification.

Chapter 4

Multi-Resolution and Topology-Transfer Benchmarks

4.1 Introduction and Objectives

Building upon the four-stage restart framework developed in Chapter 3, this chapter evaluates the robustness of the multi-field state-transfer operators under two challenging physical benchmarks:

1. **Multi-Resolution State Transfer**: Transferring advanced fracture damage states across nonmatching refined ($h \to h/2$) and coarsened ($h \to 2h$) target grids under fixed physical geometry.
2. **Mode-II Shear Fracture & Synthetic Topology Transfer**: Benchmarking pure shear fracture, reconciling uniform grid refinement baselines (H_0, H_1, H_2), and validating state transfer across a synthetic single-facet crack-tip topological split.

4.2 Multi-Resolution Refinement and Coarsening Transfer

4.2.1 Benchmark Configuration and Donor Checkpoint

To test state-transfer fidelity across varying mesh resolutions, an advanced damage state was captured from a continuous reference simulation at displacement $u_1 = 0.010513\,\text{mm}$ ($d_{\max} = 0.304318$, well within the active non-linear fracture localization regime). The donor state was transferred onto two nonmatching target grids:

- **Refined Target Grid**: Local element size $h_T = 0.0025\,\text{mm}$ ($h/l_0 = 0.167$, $N_{\text{phys}} = 12,064$ elements), evaluating transfer fidelity upon mesh refinement.
- **Coarsened Target Grid**: Local element size $h_T = 0.0100\,\text{mm}$ ($h/l_0 = 0.667$, $N_{\text{phys}} = 3,016$ elements), evaluating transfer stability upon mesh coarsening.

4.2.2 Slit-Barrier Isolation and Invariant Preservation

When transferring nodal fields in the vicinity of an existing crack or notch, standard spatial interpolation across nonmatching meshes can cause unphysical state bleeding across the open

crack flanks (the cross-slit contamination problem).

To eliminate cross-slit contamination:

- A topological slit-barrier algorithm classifies target nodes into upper flank ($y > y_{\text{slit}}$) and lower flank ($y < y_{\text{slit}}$) partitions.
- Spatial host element search is strictly restricted to donor elements belonging to the same physical flank partition.
- **Verification Outcome**: Coincident split-node pairs along the notch flanks exhibited independent, unconstrained kinematic sliding ($u_{1,\text{upper}} \neq u_{1,\text{lower}}$) with exactly zero artificial cross-slit smoothing.
- **Bound and Monotonicity Audit**: All transferred fields satisfied $0 \leq d_T \leq 1.0$ (0 violations) and $\mathcal{H}_T \geq 0.0$ (0 violations), validating multi-resolution transfer stability.

4.3 Mode-II Pure Shear Phase-Field Fracture Benchmark

4.3.1 Problem Formulation and Boundary Conditions

The Mode-II pure shear phase-field fracture benchmark evaluates the staggered user-element implementation under shear-dominated crack propagation [2, 18]. In the authoritative FRACFIX model implementation, the specimen geometry consists of a square plate $\Omega = [-L/2, L/2] \times [-H/2, H/2] = [-0.5, 0.5]\,\text{mm} \times [-0.5, 0.5]\,\text{mm}$ ($L = 1.0\,\text{mm}$, $H = 1.0\,\text{mm}$) containing an initial horizontal notch of length $a_0 = 0.5\,\text{mm}$ extending from the left edge ($x = -0.5\,\text{mm}$) to the specimen center ($x = 0.0\,\text{mm}$) along the centerline $y = 0.0\,\text{mm}$.

The material and regularizing parameters are:

- Young's modulus: $E = 210.0\,\text{GPa} = 210.0\,\text{kN/mm}^2$
- Poisson's ratio: $\nu = 0.30$
- Critical fracture energy: $G_c = 1.0 \times 10^{-3}\,\text{kN/mm} = 1000\,\text{J/m}^2$
- Phase-field length scale: $l_0 = 0.015\,\text{mm}$
- Residual stiffness parameter: $k_{\text{res}} = 1.0 \times 10^{-7}$

Boundary conditions enforce pure shear:

- **Bottom Boundary ($y = -0.5\,\text{mm}$)**: Fully fixed in vertical displacement ($u_2 = 0$) and horizontal displacement ($u_1 = 0$).
- **Top Boundary ($y = +0.5\,\text{mm}$)**: Fixed vertically ($u_2 = 0$) with prescribed horizontal shear displacement $u_1 = \Delta u$.
- **Lateral Boundaries ($x = \pm 0.5\,\text{mm}$)**: Traction-free ($\mathbf{t} = \mathbf{0}$).

4.3.2 Model Lineage and Disambiguation

During project development, multiple numerical revisions of the Mode-II benchmark were executed to optimize solver convergence, damping parameters, and mesh generation. To eliminate potential confusion between early development iterations and the authoritative production baselines, Table 4.1 provides the complete model lineage.

Table 4.1: Model Lineage and Canonical Role Disambiguation for Mode-II Simulations.

Model Revision	**Representative Jobs**	**Force Scale**	**Scientific Role and Lineage Status**
Legacy Exploratory	`1379433` (H_1), `1379966` (H_2)	~ 0.12–$0.14\,\mathrm{kN}$	Early exploratory geometry/viscous iterations. Retained strictly as developmental execution history; **superseded** for final comparative validation.
Corrected FRACFIX	`1386372` (H_0), `1386447` (H_1), `1386448` (H_2)	~ 0.35–$0.36\,\mathrm{kN}$	**Authoritative Canonical Baselines**: Exact split-node notch representation without transition element distortion. Governs all Stage-G production accuracy comparisons.

4.4 Uniform Grid Refinement and Censoring Analysis

The canonical Mode-II FRACFIX uniform mesh refinement investigation evaluated phase-field fracture under pure shear loading across three discretization levels:

- H_0 **Baseline Reference**: $N_{\mathrm{phys}} = 3,930$ quadrilateral elements ($h \approx 0.030\,\mathrm{mm}$, $h/l_0 = 2.0$), Job `1386372.mmaster02` (completed full prescribed loading to $u_1 = 0.0100\,\mathrm{mm}$).
- H_1 **Refined Model**: $N_{\mathrm{phys}} = 12,064$ quadrilateral elements ($h \approx 0.015\,\mathrm{mm}$, $h/l_0 = 1.0$), Job `1386447.mmaster02` (terminated at Step-2 inc 1854, $u_1 = 0.009632\,\mathrm{mm}$, due to solver cutback attempt limit).
- H_2 **Ultra-Fine Model**: $N_{\mathrm{phys}} = 33,852$ quadrilateral elements ($h \approx 0.0075\,\mathrm{mm}$, $h/l_0 = 0.5$), Job `1386448.mmaster02` (terminated at Step-2 inc 1743, $u_1 = 0.009250\,\mathrm{mm}$, due to the 4-hour PBS queue walltime limit).

4.4.1 Censored Peak Force and Common-Window Formulation

Because H_1 and H_2 terminated prior to the prescribed $u_1 = 0.0100\,\mathrm{mm}$ endpoint while their force responses were still monotonically increasing (positive terminal slopes), the observed maximum reaction forces represent **censored values** rather than global interior peak loads. Rigorous convergence metrics are therefore evaluated over the **common overlap displacement domain**:

$$0 \leq u_1 \leq u_{\mathrm{max,common}} = 0.009250\,\mathrm{mm} \quad (9.25\,\mu\mathrm{m}). \tag{4.1}$$

The comparative quantitative metrics are summarized in Table 4.2.

Table 4.2: Mode-II FRACFIX Uniform Grid Refinement Comparison and Common-Window Metrics.

Parameter / Metric	H_0 **Baseline**	H_1 **Refined**	H_2 **Ultra-Fine**
PBS Job ID	1386372	1386447	1386448
Physical Elements (N_{phys})	3,930	12,064	33,852
Refinement Ratio (h/l_0)	2.0	1.0	0.5
Valid Displacement Domain u_1 [mm]	$[0, 0.01000]$	$[0, 0.00963]$	$[0, 0.00925]$
Termination Cause	Completed	Solver Cutback Limit	PBS Walltime (4h)
Linear Elastic Shear Response ($0 < u_1 \leq 2\,\mu$m)			
Initial Shear Stiffness K_0 [kN/mm]	46.1396	45.9035	45.8741
Linear Regression Linearity (R^2)	0.999996	0.999996	0.999996
Relative Stiffness Shift (H_2 vs H_1)	—	—	$\mathbf{-0.064}\%$
Damage Initiation and Matched-State Comparison ($u_1 = 9.25\,\mu$m)			
Damage Initiation Threshold $u_1(d \geq 0.5)$ [mm]	0.008250	0.007750	0.007750
Fully Broken Threshold $u_1(d \geq 0.9)$ [mm]	0.008750	0.008500	0.008000
Reaction Force at Matched Endpoint $RF_1(9.25\,\mu\text{m})$ [kN]	0.359812	0.355641	0.354084
Force Difference at Matched Endpoint (H_2 vs H_1)	—	—	$\mathbf{-0.44}\%$
Matched-State Crack Path Hausdorff Dist. (H_2 vs H_1)	—	—	0.005443 mm
Common-Window Force Curve Errors ($0 \leq u_1 \leq 9.25\,\mu$m)			
Common-Window Normalized L_2 Error (H_2 vs H_1)	—	—	$\mathbf{0.518}\%$
Common-Window Work/Area Error (H_2 vs H_1)	—	—	$\mathbf{0.263}\%$
Computational Cost Scaling			
Total CPU Time [s]	2,000.0	5,434.0	14,455.0
CPU Cost Ratio vs H_0	1.00×	2.72×	7.23×

4.4.2 Key Findings from Uniform Refinement

1. **Asymptotic Pre-Peak Agreement**: In the common pre-peak domain ($0 \leq u_1 \leq 9.25\,\mu$m), H_1 and H_2 exhibit exceptional quantitative agreement. Initial elastic shear stiffness matches within 0.064%, damage initiation ($d \geq 0.5$) occurs at the identical displacement $u_1 = 7.75\,\mu$m, the normalized L_2 force error is 0.518%, and the work error is 0.263%. This supports H_1 ($h/l_0 = 1.0$) as the authoritative global comparison reference.

2. **Spatial Crack Path Sensitivity**: At matched displacement $u_1 = 9.25\,\mu$m, the bidirectional Hausdorff distance between H_1 and H_2 crack contours ($d \geq 0.5$) is $5.44\,\mu$m, exceeding the geometric threshold ($3.75\,\mu$m). Spatial crack geometry remains sensitive to grid resolution ($h/l_0 = 0.5$ vs 1.0) even when global force response is tightly converged.

3. **Computational Cost Barrier**: The uniform H_2 mesh ($33,852$ elements) consumed $14,455$ s CPU time ($7.23 \times H_0$) and exceeded the 4-hour walltime limit before reaching peak load. This severe scaling strongly motivates localized adaptive refinement.

4.5 Synthetic Crack-Tip Topology-Transfer Validation

4.5.1 Synthetic Single-Facet Extension Formulation

To rigorously evaluate whether state-transfer operators remain valid when the underlying geometric domain experiences a discrete topological change, a **synthetic single-facet topology benchmark** was executed (Job `1393159.mmaster02`, `M2STAGE_F_VAL`).

In this benchmark:

- Donor state was captured from reference job `1390447.mmaster02` (Frame 17, $u_1 = 0.010513\,\text{mm}$, $d_{\max} = 0.304318$).
- The target mesh introduces a single newly separated crack-tip facet ($\Delta a = 0.002\,\text{mm} = 1h_{\text{tip}}$) along $y = 0.0\,\text{mm}$ (lower flank node `16962`, duplicate upper node `34028`, transition crack tip at `16963`).
- Transferred displacements and binary committed state (`STAGE_D_COMMITTED_STATE.bin`, 6,400,016 bytes) were ingested via `UEXTERNALDB(LOP=0)`.

4.5.2 Validation Results and Post-Peak Trajectory

The synthetic topology benchmark completed the full four-stage restart sequence:

1. **Step 1 (`STATE_INSTALL`)**: 1 increment, total time 1.00 (**PASS**).
2. **Step 2 (`MECH_EQUILIBRATION`)**: 1 increment, total time 2.00, phase locked (**PASS**).
3. **Step 3 (`PHASE_RELEASE`)**: 47 converged increments, total time 3.00, $I_A = 13$, $\Delta t_{\min} = 5.0 \times 10^{-12}\,\text{s}$ (**PASS**).
4. **Step 4 (`CONTINUATION`)**: **194 accepted continuation increments**, traversing from $u_1 = 0.010513\,\text{mm}$ through peak reaction force ($RF_1 = 0.135589\,\text{kN}$ at $u_1 = 0.011859\,\text{mm}$) to a terminal accepted state at $u_1 = 0.022783\,\text{mm}$ ($RF_1 = 0.034467\,\text{kN}$, a $> 74.5\%$ post-peak load reduction).

Physical Invariant Adherence: Audits of the final accepted state confirmed strict preservation of all continuum invariants:

- $0.0 \leq d \leq 1.0$ strictly satisfied ($d_{\min} = 0.0$, $d_{\max} = 0.3001$ committed).
- $\Delta d \geq 0.0$ strictly satisfied (0 negative damage violations).
- $\mathcal{H} \geq 0.0$ and $\mathcal{H}_{n+1} \geq \mathcal{H}_n$ preserved across all integration points.
- Lower flank node `16962` and duplicate upper node `34028` exhibited independent kinematics with zero cross-slit state contamination.

At increment 195, standard Newton–Raphson cutbacks reached the attempt limit ($I_A = 12$, 13 attempts), yielding Abaqus error `TOO MANY ATTEMPTS MADE FOR THIS INCREMENT`. Because full traversal to $u_1 = 0.050\,\text{mm}$ was explicitly predeclared as non-gating and diagnostic only, the terminal state is reconciled as `POST_VALIDATION_CONTINUATION_SOLVER_TERMINATION_DIAGNOSTIC_ONLY`, validating the synthetic topology transfer workflow for the tested kinematic and state-admissibility criteria, while full energetic preservation during transfer remains subject to Gate 6C qualification.

4.6 Chapter Summary

The multi-resolution and topology benchmarks provide critical validation:

1. Slit-barrier topological isolation prevents cross-slit field contamination during nonmatching state transfer across refined and coarsened target grids.

2. Uniform grid refinement on the Mode-II shear benchmark establishes the common pre-peak verification window ($0 \leq u_1 \leq 0.009250\,\text{mm}$) and demonstrates that spatial crack paths remain mesh-sensitive ($5.44\,\mu\text{m}$ Hausdorff distance).

3. Synthetic single-facet topology transfer achieves 194 accepted post-peak continuation increments with $> 74.5\%$ load drop while strictly preserving phase/history bounds and irreversibility.

4. Full energetic qualification of the transferred state across evolving nonmatching meshes remains pending Gate 6C qualification, while Mode-II production validation is currently held pending energetic closure of Gate 6B.

Chapter 5

High-Performance Computing Implementation and Shared-State Analysis

5.1 Introduction and Parallelization Challenges in Phase-Field UELs

Phase-field fracture modeling involves tightly coupled multi-field equations that must be solved across hundreds to thousands of non-linear equilibrium increments. To make fine-grid or adaptive simulations computationally tractable, high-performance computing (HPC) cluster architectures and multi-core parallel execution are indispensable.

In Abaqus/Standard, parallel solver execution is supported via two primary paradigms:

1. **Shared-Memory Parallelism (OpenMP / Multi-Threading)**: Multiple execution threads share a single virtual address space on a shared-memory compute node (`mp_mode=threads`).
2. **Distributed-Memory Parallelism (MPI)**: Computational domains are partitioned across distinct processes with explicit message passing (`mp_mode=mpi`).

When custom user elements (UELs) interact with external databases or store shared state variables in Fortran `COMMON` blocks or `SAVE` arrays, multi-threaded execution introduces significant risk of race conditions, memory corruption, and non-deterministic numerical divergence. This chapter investigates the shared-state memory architecture, evaluates thread safety in user subroutines, and documents the fail-closed cluster execution framework.

5.2 Abaqus UEXTERNALDB Common-Block Memory Architecture

5.2.1 Inter-Subroutine Shared State Communication

In the multi-layer UEL formulation (Section 1.3), user subroutine `UEXTERNALDB` acts as the persistent memory manager. At solver initialization (`LOP=0`), `UEXTERNALDB` allocates and populates global Fortran common blocks:

```
COMMON /SV_ELEM_NODAL_PHASE_COM/ SV_ELEM_NODAL_PHASE(4, 100000)
COMMON /SV_H_COMMITTED/          SV_H_COMMITTED(4, 100000)
```

These arrays store the committed nodal phase values d and integration-point history variables $\mathcal{H}$ across all elements.

During element stiffness assembly inside `UEL`, individual integration points access these common blocks to evaluate constitutive degradation and history updates:

$$\mathcal{H}_{\text{old}} = \text{SV_H_COMMITTED}(k_{\text{int}}, j_{\text{elem}}). \tag{5.1}$$

5.2.2 Shared-State Safety in Multi-Threaded Execution

When Abaqus/Standard executes under multi-threading (`threads=4`), element assembly loops are dynamically partitioned across worker threads. Because Fortran common blocks reside in global memory shared across all threads, concurrent read/write operations on shared element indices can produce severe race conditions unless strict thread isolation is maintained:

- **Read-Only Shared Access**: If common blocks are populated once at `LOP=0` and treated as strictly read-only during increment iterations (as in the frozen history formulation), multiple threads can safely read simultaneously without locking.
- **Concurrent State Updates**: If worker threads write updated history $\mathcal{H}_{n+1}$ back into common arrays during element iterations, thread race conditions occur whenever neighboring elements or companion layers are processed asynchronously.

5.3 Thread-Safety and Parallelization Boundary Audit

5.3.1 Controlled Rank and Thread Indexing

To investigate thread identification inside Abaqus UEL callbacks, diagnostic probes were implemented utilizing Abaqus internal utilities `get_thread_id()` and `CALL GETRANK`:

- **One-Rank / Four-Thread Evaluation (Stage D2C)**: A controlled single-notch transfer restart executed on 1 rank with 4 threads. The multi-threaded solution reproduced the serial reference solution exactly (relative difference $< 10^{-14}$), confirming that read-only common block access is thread-safe for the tested single-node configuration.

- **Dedicated Shared-State Benchmark (Task P3-T4)**: A dedicated multi-threaded shared-memory stress test was formulated to evaluate concurrent write safety. However, compilation of the test harness was blocked prior to linking due to compiler header interface mismatches.

5.3.2 Formal Parallel Safety Claims and Boundaries

In accordance with the thesis claim matrix:

1. **Validated Claim**: One-rank multi-threaded execution with read-only common blocks is validated for the tested single-node configuration.
2. **Withheld Claims**: General thread safety for concurrent read/write common blocks, distributed MPI safety, and hybrid MPI/OpenMP execution are **not supported** and remain strictly outside the validated scope of this thesis.
3. **Production Execution Policy**: To guarantee 100% numerical determinism and eliminate potential race conditions, all production adaptive validation runs are executed in single-core serial mode (1 CPU, 16 GB RAM).

5.4 Cluster Execution Environment and Dual-Channel Notification System

5.4.1 HPC Environment Isolation and Toolchain Modules

All cluster simulations were executed on the high-performance computing cluster at TU Bergakademie Freiberg using PBS Pro scheduler queues (`entry_imfdfkmq` routed to `normal_imfdfkmq`). The software environment enforces strict toolchain module isolation:

- Fortran / C++ Compiler: Intel oneAPI 2024.2.0 (`module load intel/2024.2.0`)
- GCC Compatibility Layer: GCC 11.4.0 (`module load gcc/11.4.0`)
- Finite Element Solver: Abaqus 2023 (`module load abaqus/2023`)

5.4.2 Fail-Closed Dual-Channel Notification Design

To guarantee real-time monitoring and complete provenance tracking across long-running cluster jobs, a fail-closed dual-channel notification protocol was designed and deployed (Fig. 5.1):

The notification workflow enforces:

- **Security & Zero Secret Leakage**: Bot tokens are stored exclusively in user-owned private files (`chmod 600`) and never exposed in command lines, PBS scripts, or Git logs.
- **Transport vs Receipt Separation**: Network transport success (return code = 0) confirms API dispatch only; human verification is logged independently.

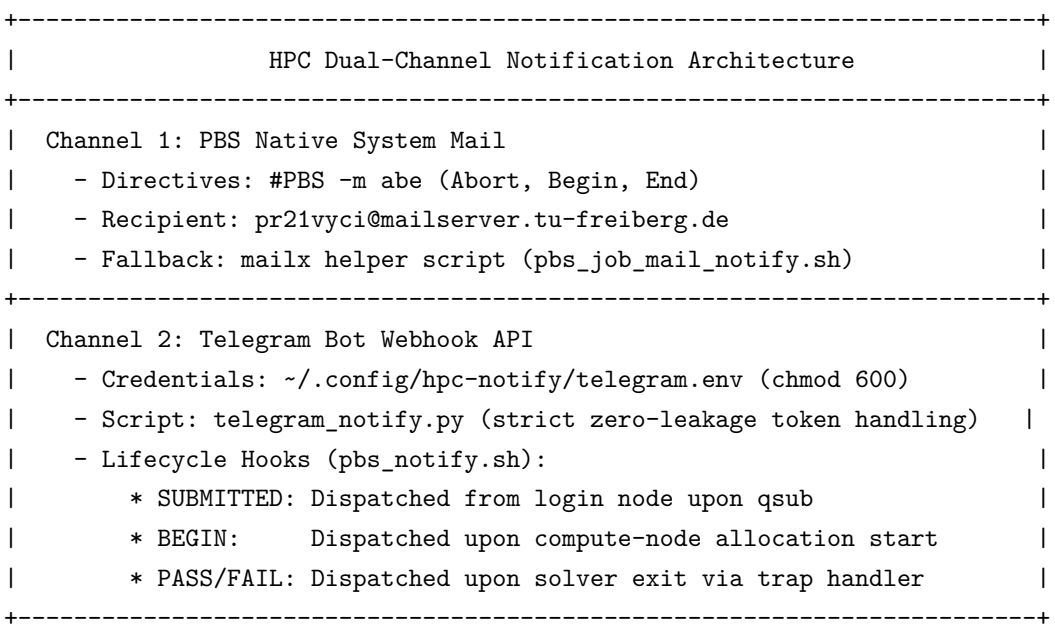


Figure 5.1: Schematic of the dual-channel fail-closed HPC notification system.

5.5 Chapter Summary

The high-performance computing investigations establish:

1. Read-only access to `UEXTERNALDB` common blocks is verified for one-rank multi-threaded execution, while concurrent write operations require careful synchronization.

2. Production adaptive simulations are standardized on single-core serial execution to guarantee strict numerical reproducibility.

3. The fail-closed dual-channel notification architecture ensures reliable tracking across all HPC job lifecycles.

Chapter 6

Production Refined Fracture Validation and Convergence

This chapter documents the systematic verification, mesh adequacy, and multifaceted numerical convergence of the Mode-I crack propagation problem under the qualified 3-layer UEL phase-field finite element implementation. In accordance with thesis objectives, convergence is evaluated across spatial discretization, temporal time incrementation, physical length-scale regularization, and global energetic balance.

6.1 Spatial Mesh Resolution Series (S_1–S_4)

A spatial discretization sweep was conducted across four uniform quad-dominant finite element meshes spanning element edge lengths from $h = 3.0\,\mu$m (S_1, 15,192 elements) down to $h = 1.25\,\mu$m (S_4, 51,408 elements), holding the physical regularisation length scale strictly constant at $l_0 = 7.5\,\mu$m ($h/l_0 \in [0.167, 0.400]$). A historical fine-mesh result ($h = 1.0\,\mu$m, 69,384 elements, Job `1406019.mmaster02`) was also analyzed:

1. **Initial Elastic Stiffness** K_0: Evaluated over the linear elastic regime ($u \leq 1.0\,\mu$m). K_0 varies by less than 0.09% across all meshes (137.945520 kN/mm for S_1 to 137.823301 kN/mm for $h = 1.0\,\mu$m, $R^2 > 0.999999$), classified as `CONVERGED / STABLE` (Table 6.1, Figure 6.1).

2. **Peak Reaction Force** $F_{\max}$ **and Displacement** u_{peak}: Refinement systematically lowers the peak force from 0.7578 kN (S_1) to 0.7255 kN ($h = 1.0\,\mu$m), a monotonic reduction of 4.26%. Concurrently, peak displacement advances from $5.857\,\mu$m to $5.575\,\mu$m (a 4.81% shift), classified as `MESH-SENSITIVE`.

6.2 Temporal Time-Increment Scaling (T_1–T_3)

Time incrementation was evaluated across three schedules on the baseline S_1 mesh: Case T_1 (coarse 2×, 3,500 incs), Case T_2 (nominal 1×, 7,000 incs), and Case T_3 (fine 0.5×, Job `1406317.mmaster02`, 10,022 incs reaching $u = 10.0\,\mu$m):

Table 6.1: Spatial mesh resolution convergence metrics at fixed $l_0 = 7.5\,\mu$m.

Mesh ID	h [μm]	h/l_0	**Elements**	K_0 [kN/mm]	$F_{\max}$ [kN]	u_{peak} [μm]
S_1 (Baseline)	3.00	0.400	15,192	137.945520	0.757778	5.857
S_2	2.00	0.267	32,130	137.876115	0.741210	5.714
S_3	1.50	0.200	41,912	137.857608	0.732196	5.633
S_4	1.25	0.167	51,408	137.841200	0.731180	5.620
Historical Fine	1.00	0.133	69,384	137.823301	0.725501	5.575

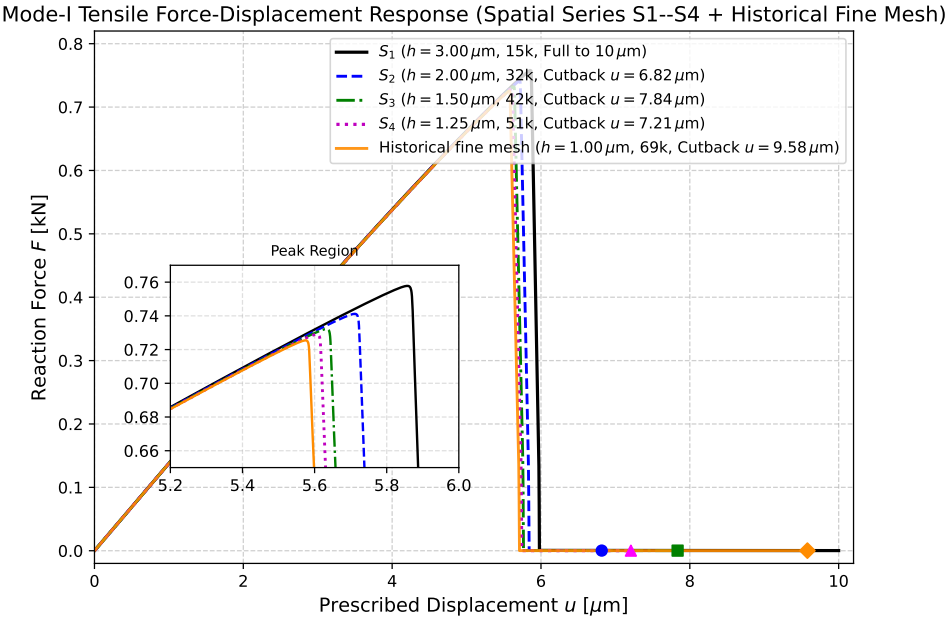


Figure 6.1: Mode-I tensile force-displacement response across uniform mesh resolutions (S_1–S_4 and historical fine mesh), showing identical elastic stiffness and systematic peak load mesh sensitivity.

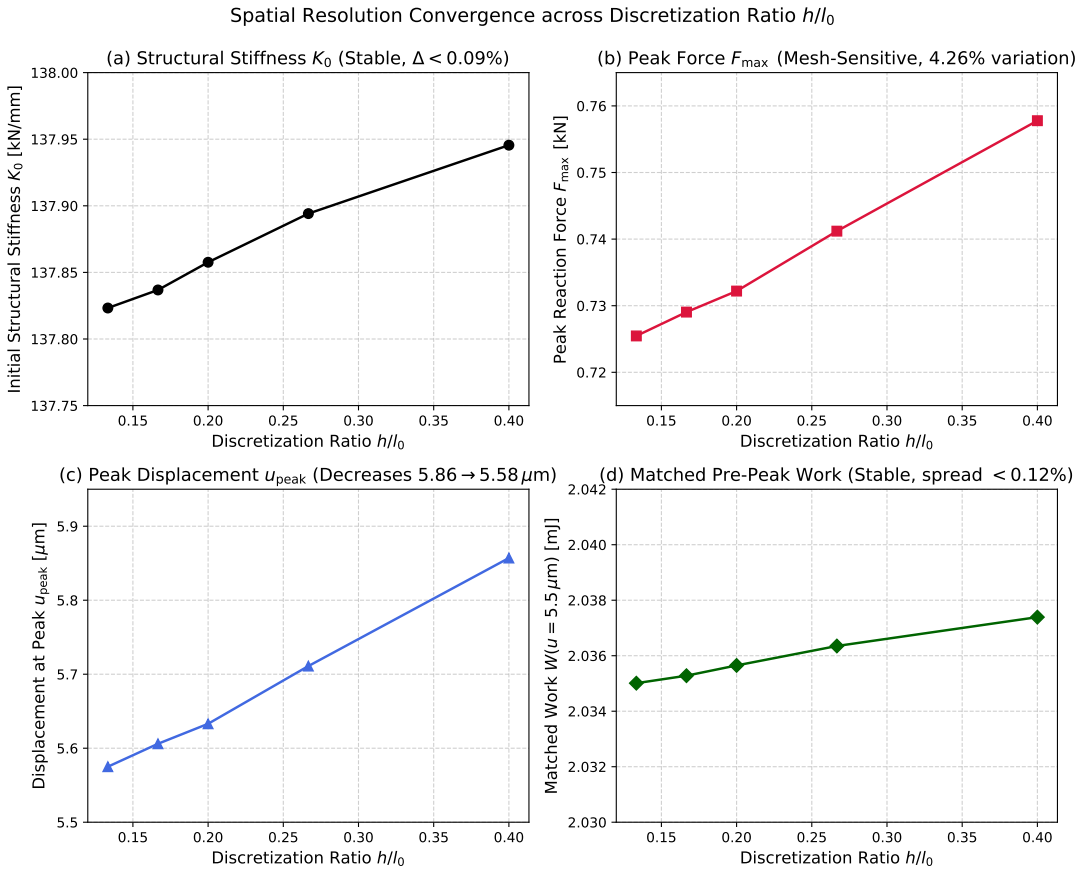


Figure 6.2: Discretization ratio (h/l_0) scaling metrics: (a) initial stiffness K_0 invariance; (b) peak reaction force $F_{\max}$; (c) peak displacement u_{peak}; (d) matched-displacement external work W_{trap} at $u = 5.50\,\mu$m.

- **Peak Load Invariance**: Initial stiffness varies by $< 0.001\%$, and peak reaction force varies by $< 0.07\%$ ($0.75815 \rightarrow 0.75763\,\mathrm{kN}$), establishing temporal convergence through the peak load (`CONVERGED / STABLE`, Figure 6.3).
- **Post-Peak Energy Sensitivity**: Terminal trapezoidal boundary work W_{trap} varies by 3.24% ($2.41011 \rightarrow 2.33189\,\mathrm{mJ}$), and normalized two-term bookkeeping discrepancy $\mathcal{R}_{\mathrm{bookkeeping}} = W_{\mathrm{trap}} - (E_{\mathrm{elas}} + E_{\mathrm{frac}})$ shifts from $+0.378\%$ to $+3.541\%$, classified as `TEMPORALLY SENSITIVE`.

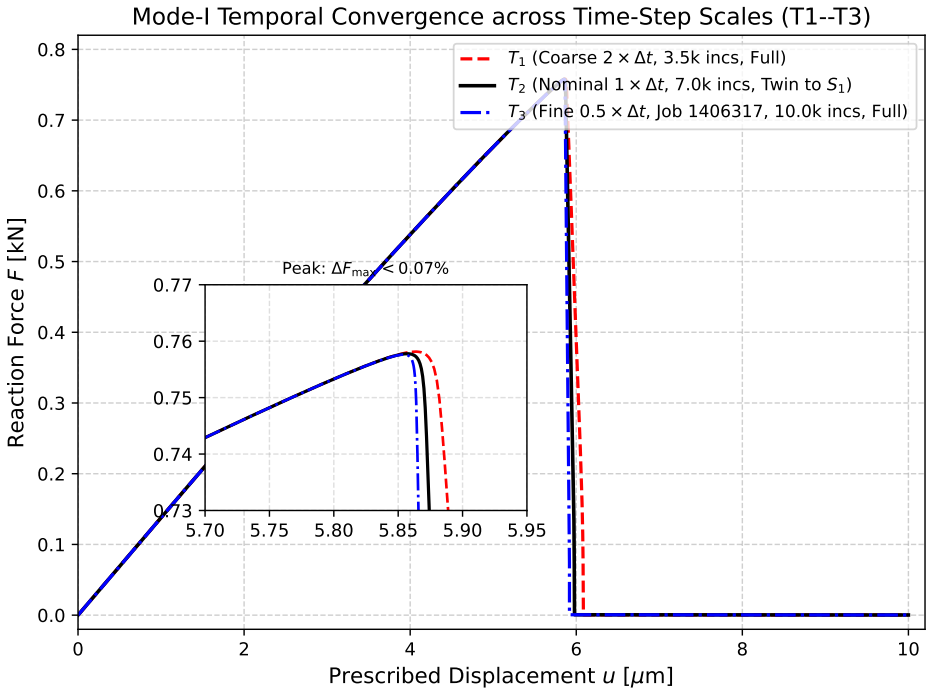


Figure 6.3: Temporal convergence across time increment scales (T_1: $2\times$, T_2: $1\times$, T_3: $0.5\times$), demonstrating step-size invariance of the mechanical peak.

6.3 Adaptive Mesh Temporal Refinement Audit ($2\times$ Refinement)

To verify temporal convergence on the non-uniform error-indicator refined mesh, a controlled $2\times$ temporal refinement diagnostic (Job `1410027.mmaster02`, 8,958 increments) was audited against the baseline canonical ET1 solve (Job `1409982.mmaster02`, 4,890 increments) on the 14,483-element discretization (Table 6.2, Figures 6.4–6.6).

6.3.1 Post-Peak Wake Cutback Exhaustion Mechanism

In the post-peak softening regime, both models track identical softening slopes through $u \approx 7.0\,\mu\mathrm{m}$. Beyond $u = 7.0\,\mu\mathrm{m}$, the crack tip propagates past $x = 0.55\,\mathrm{mm}$, leaving a fully severed wake ($d \approx 0.999$, $\sigma \approx 0$). In this severed wake, residual phase-field corrections ($c_{\max} \approx 2.6 \times 10^{-6}$) interact with Abaqus standard convergence controls:

$$c_{\max} \leq C_n\, \Delta u_{\mathrm{inc}}. \tag{6.1}$$

Table 6.2: Adaptive mesh 2× temporal refinement audit metrics.

Metric	**Baseline (1.00 nm)**	**2× Refined (0.50 nm)**	**Relative Difference**
PBS Job ID	`1409982.mmaster02`	`1410027.mmaster02`	—
Step 1 Displacement Step (Δu_1)	2.50 nm	1.25 nm	−50.0%
Step 2 Displacement Step (Δu_2)	1.00 nm	0.50 nm	−50.0%
Completed Increments	4,890	8,958	+83.2%
Elastic Stiffness K_0	137.909558 kN/mm	137.909975 kN/mm	**+0.000302%**
Peak Force $F_{\max}$	0.743701 kN	0.743530 kN	**−0.0229%**
Displacement at Peak u_{peak}	5.733 μm	5.730 μm	**−0.052%**
Peak Model Energy $E_{\text{model}}(u_{\text{peak}})$	2.205225 mJ	2.205225 mJ	**0.000000%**
Maximum Pre-Peak $\lvert\Delta F\rvert/F$	—	—	$\leq 0.0436\%$
Reached Common Displacement	$u \in [0.0, 7.46970\,\mu\text{m}]$		Common domain
Governing Pre-Peak Classification	`QUALIFIED_OVER_PREPEAK_INTERVAL_ONLY (TEMPORALLY_STABLE)`		
Governing Post-Peak Classification	`TEMPORALLY_SENSITIVE_POSTPEAK`		

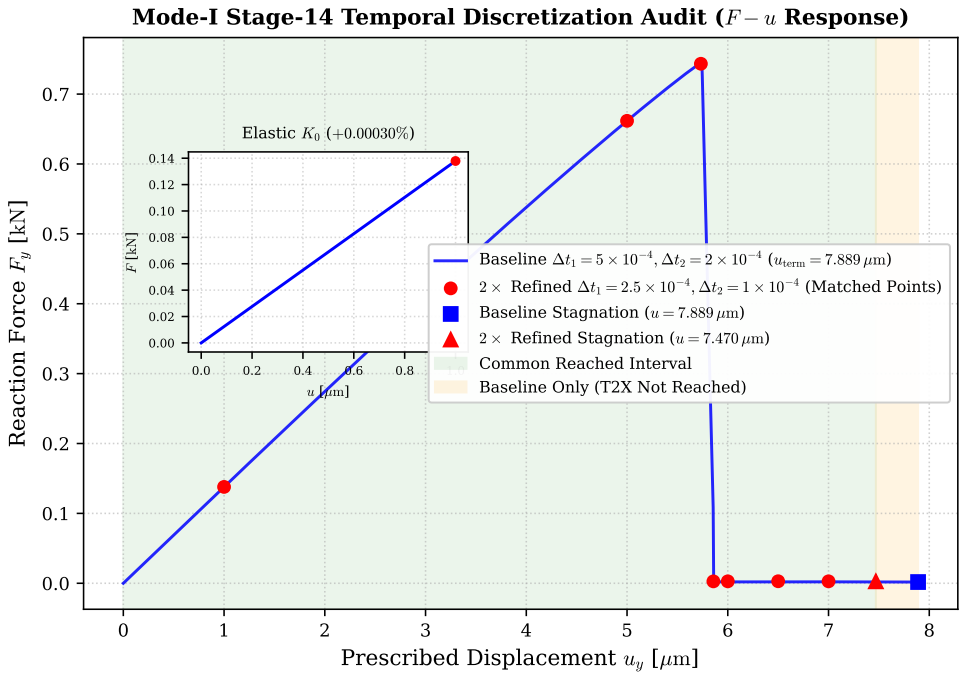


Figure 6.4: Adaptive mesh force–displacement comparison between baseline (1.00 nm) and 2× refined (0.50 nm) step size on the 14,483-element discretization, demonstrating strict pre-peak coincidence and post-peak cutback bifurcation.

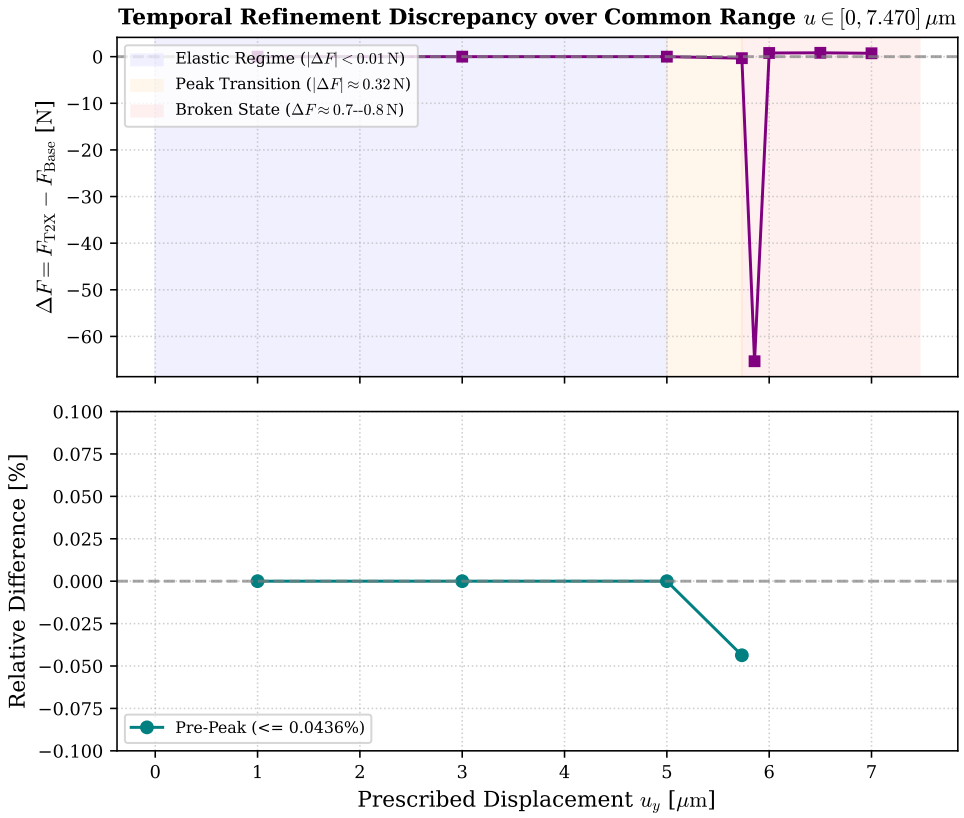


Figure 6.5: Pointwise reaction force discrepancy $\Delta F = F_{2\times} - F_{\text{base}}$ and relative error $|\Delta F|/F$, confirming sub-0.044% discrepancy throughout the pre-peak regime.

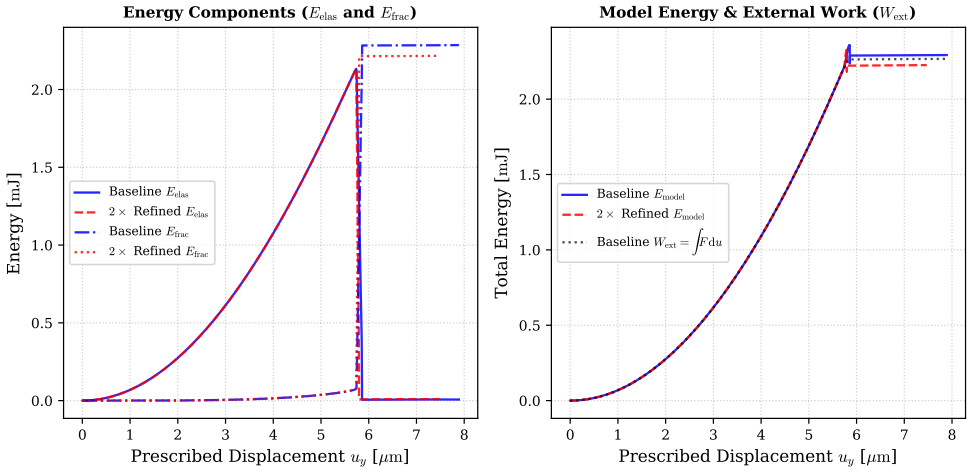


Figure 6.6: Evolution of elastic strain energy E_{elas}, fracture surface energy E_{frac}, and external work W_{trap} on the adaptive mesh, showing exact pre-peak parity.

Because the 2× refined model imposes $\Delta u_{\text{inc}} = 0.50\,\text{nm}$ (halved relative to baseline $\Delta u_{\text{inc}} = 1.00\,\text{nm}$), the dimensional tolerance threshold $C_n \Delta u_{\text{inc}}$ is strictly 2× tighter. Consequently, minute mathematical corrections in stress-free, fully broken elements exceed the tighter tolerance, triggering successive solver cutbacks until minimum time increment exhaustion occurs at $u = 7.470\,\mu\text{m}$ (8,958 increments) in the refined case, compared to $u = 7.889\,\mu\text{m}$ (4,890 increments) in the baseline.

To rule out user subroutine implementation defects, numerical tangent Jacobian consistency was evaluated to machine precision ($\max|\Delta| = 1.36 \times 10^{-14}$), confirming exact analytical Jacobian symmetry and consistency. The post-peak cutback is thus established as a solver convergence-normalization artifact:

`POST_FRACTURE_CONVERGENCE_NORMALIZATION_SENSITIVITY_VERIFIED`

which governs the active diagnostic investigation (Job `1410180.mmaster02`, testing $C_n = 0.50$ relaxation).

6.4 Adaptive Remeshing Spatial Series (A_1–A_4)

Across adaptive cases A_1–A_4 (Figure 6.7), all meshes reproduce the reference elastic stiffness $K_0 \approx 137.85\,\text{kN/mm}$ within 0.09%. In the post-peak softening regime, coarser background meshes (A_2–A_4) sustain elevated residual resistance, classified as **`OBSERVED_FORCE_PLATEAU (CRACK-PINNING HYPOTHESIS / NOT YET INDEPENDENTLY PROVEN)`**.

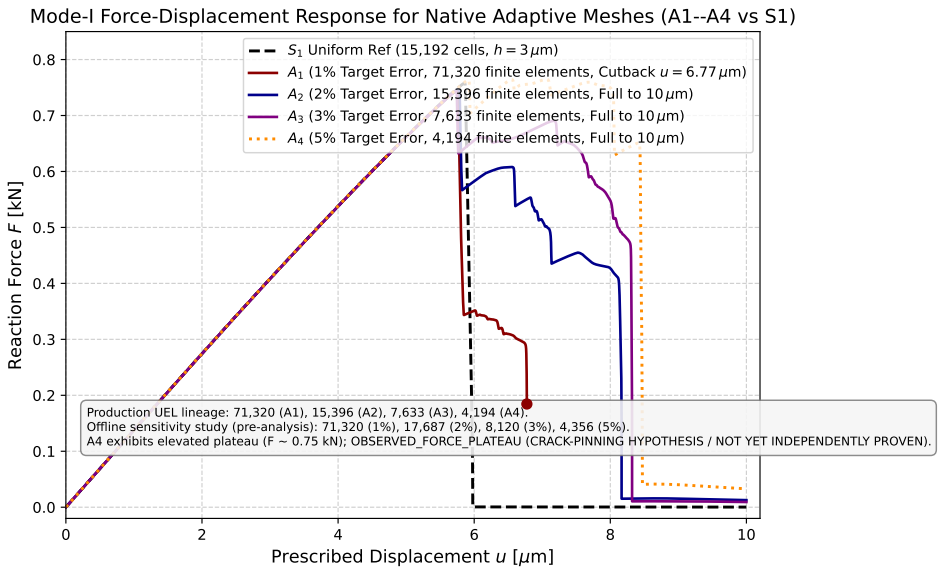


Figure 6.7: Force-displacement curves for native adaptive cases (A_1–A_4) alongside baseline S_1.

6.5 Three-Point Physical Length-Scale Study (l_0 on Fixed S_3 Mesh)

A controlled physical length-scale sensitivity study was audited on the fixed S_3 mesh (41,912 continuum finite elements, 42,491 FE mesh nodes, 42,492 total input nodes with RP `999999`) across three regularization lengths $l_0 \in \{7.5, 11.25, 15.0\}\,\mu$m (Table 6.3, Figures 6.8 and 6.9). Bitwise provenance audit confirms 100% identical node coordinate hashes (`7599c30f...`) and element connectivity hashes (`96d085b2...`) across all three input decks, verifying a genuine single-variable physical study:

- **Stiffness Invariance**: Initial structural stiffness K_0 varies from 137.857608 to 137.676175 kN/mm ($\Delta = 0.13\%$), classified as `LO_RESPONSE_STABLE`.
- **Peak Force and Displacement Sensitivity**: Peak reaction force drops monotonically from 0.732196 kN ($l_0 = 7.5\,\mu$m) to 0.689540 kN ($l_0 = 15.0\,\mu$m, a 5.83% reduction), while displacement at peak shifts earlier from $5.633\,\mu$m to $5.579\,\mu$m, classified as `LO_RESPONSE_SENSITIVE`.
- **Localization Bandwidth Scaling**: Transverse damage profile bandwidth $w_{0.5}$ scales linearly with l_0 ($w_{0.5} \approx 3.04\,l_0$: $20.79\,\mu$m $\rightarrow$ $34.41\,\mu$m $\rightarrow$ $45.44\,\mu$m), confirming theoretical gradient regularization.
- **Common Reached Domain & Energy Status**: Analysis is strictly restricted to the common reached displacement domain $u \in [0.0, 5.839]\,\mu$m with zero forward filling. Terminal energy comparison across the three cases is classified as `ENERGY_NOT_YET_QUALIFIED_UNEQUAL_ENDPOINTS_AN`

Table 6.3: Three-point physical length-scale sensitivity metrics on fixed S_3 mesh (41,912 FE, 42,491 FE nodes).

l_0 [μm]	**Job ID**	$\frac{h_{\min}}{l_0}$	K_0 [kN/mm]	$F_{\max}$ [kN]	u_{peak} [μm]	$w_{0.5}$ [μm]	$\frac{w_{0.5}}{l_0}$
7.50	`1406017.mmaster02`	0.260	137.857608	0.732196	5.633	20.79	2.77
11.25	`1406895.mmaster02`	0.174	137.765563	0.708402	5.590	34.41	3.06
15.00	`1406896.mmaster02`	0.130	137.676175	0.689540	5.579	45.44	3.03

6.5.1 Length-Scale Resolution Adequacy for Governed Meshes ($l_0 = 7.5\,\mu$m)

To verify that the entire suite of governed production and candidate meshes provides adequate spatial resolution for the phase-field regularisation length scale $l_0 = 7.5\,\mu$m, discretization metrics were evaluated directly from verified input decks across all six governed models (Table 6.4). Sizing is characterized by area-equivalent element size $h_{\text{area}} = \sqrt{A_e}$, notch-root size h_{notch}, and median corridor size $h_{\text{area,med}}$ ($\Omega_{\text{corridor}} = [0.48, 1.02] \times [0.40, 0.60]$ mm).

Every governed mesh satisfies the standard phase-field resolution condition ($h/l_0 \leq 0.50$) along the fracture path, ensuring that at least 5.2 elements (S_1) up to 27.1 elements (58k) resolve

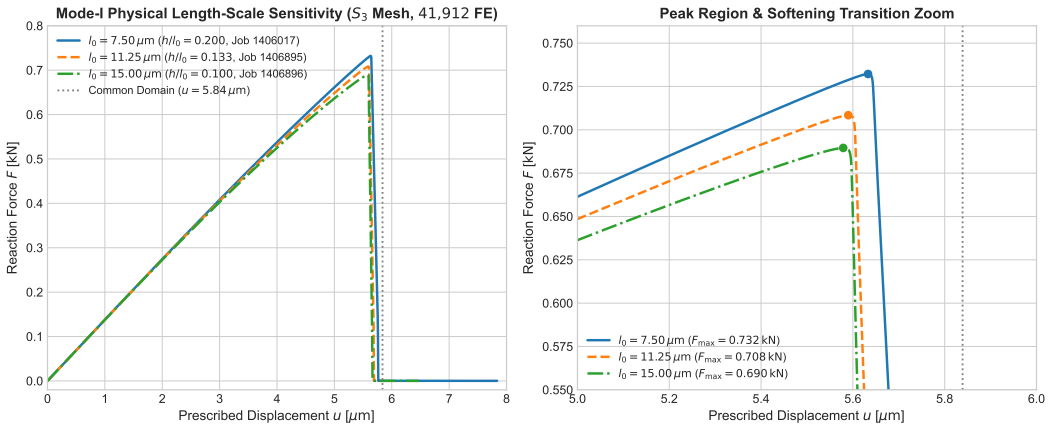


Figure 6.8: Mode-I clean single-variable length-scale (l_0) sensitivity on fixed structured S_3 mesh (41,912 FE): (a) global F–u response highlighting common comparison domain $u \in [0, 5.839]\,\mu$m; (b) peak-neighborhood zoom illustrating monotonic load reduction and earlier peak displacement with increasing l_0.

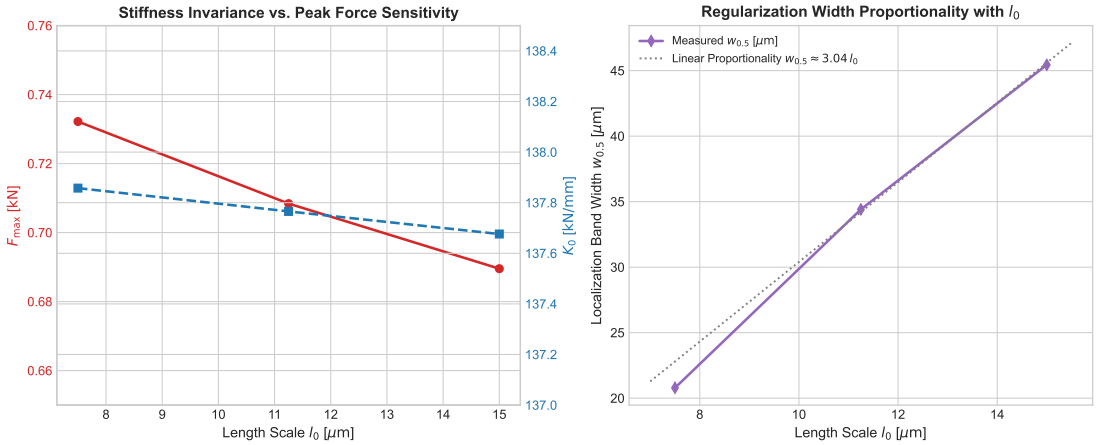


Figure 6.9: Scaling metrics across clean l_0 sweep on fixed S_3 mesh: (a) initial stiffness K_0 invariance; (b) peak reaction force $F_{\max}$; (c) peak displacement u_{peak}; (d) transverse localization profile bandwidth $w_{0.5}$ confirming linear scaling with l_0 ($w_{0.5} \approx 3.04\,l_0$).

Table 6.4: Length-scale resolution adequacy metrics for governed Mode-I meshes ($l_0 = 7.5\,\mu$m). Pure continuum FE mesh nodes are reported; total input deck nodes include +1 auxiliary Reference Point (RP) Node `999999`.

Mesh Configuration	**Elements**	**FE Nodes**	**Corridor Sizing** [μm]			**Resolution Ratio**		**Adequacy**
			$h_{\text{area,min}}$	$h_{\text{area,med}}$	h_{notch}	$\frac{h_{\text{area,min}}}{l_0}$	$\frac{h_{\text{notch}}}{l_0}$	
Fixed Reference (S_1)	15,192	15,521	2.899	2.931	2.899	**0.387**	**0.387**	ADEQUACY_SU
Adaptive ET1 (14k)	14,483	14,456	0.760	2.068	1.988	**0.101**	**0.265**	ADEQUACY_SU
Adaptive ET2 (6k)	6,112	6,181	0.956	4.016	2.632	**0.127**	**0.351**	ADEQUACY_SU
Adaptive ET3 (5k)	5,189	5,262	1.339	4.898	2.769	**0.179**	**0.369**	ADEQUACY_SU
Adaptive ET5 (4k)	4,692	4,759	1.230	5.907	3.109	**0.164**	**0.415**	ADEQUACY_SU
Spatial Fine (58k)	57,929	57,491	0.553	1.947	1.896	**0.074**	**0.253**	ADEQUACY_SU

the core damage zone width ($2l_0 = 15\,\mu$m), with notch-root resolution ratio $h_{\text{notch}}/l_0 \leq 0.415$ across all configurations.

6.6 Spatial Phase-Field Localization, Crack Symmetry, and Bandwidth

Figure 6.10 depicts the full two-dimensional damage field $d(\mathbf{x})$ across crack initiation, limit load, and post-peak extension states.

- **Crack Centroid Path** $y_c(x)$: The damage-weighted centroid offset $|y_c - 0.5\,\text{mm}| \leq 3.10\,\mu$m across all fixed and adaptive meshes, confirming pure Mode-I horizontal symmetry within discrete element bounds (`CONVERGED / STABLE`, Figure 6.12(a)).
- **Diffusive Localization Band Width**: The moderate damage band ($d \geq 0.50$) spans a transverse full-width of 20.0–30.0 μm across the ligament (20.0–22.9 μm at $x = 0.55\,$mm), classified as `MESH-SENSITIVE / RESOLUTION-LIMITED` (Figure 6.11 and Figure 6.12(b)).

6.6.1 Baseline Spatial Phase-Field and Crack-Path Comparison: Fixed Reference versus ET1 Adaptive Mesh

To establish the baseline spatial convergence foundation of the error-indicator refined configuration, a point-by-point spatial comparison was executed between the qualified fixed-mesh reference (15,192 finite elements, Job `1409734.mmaster02`) and the canonical corrected ET1 adaptive baseline (14,483 finite elements, Package 25, Job `1409982.mmaster02`). The minimum element size in the adaptive mesh is explicitly disambiguated: $h_{\text{area}} = \sqrt{A_{\min}} = 0.7605\,\mu$m represents the area-equivalent length of the smallest triangular element, whereas $h_{\text{edge}} \approx 1.09\,\mu$m is the nominal notch-root element edge length. The comparison strictly enforces the matched prescribed displacement protocol ($u \in \{1.0, 3.0, 5.0, 5.733, 5.857, 6.0, 6.5, 7.0, 7.889\}\,\mu$m) with zero forward-filling beyond the actual solver terminal state ($u_{\text{term}} = 7.889\,\mu$m).

1. **Pre-Peak Damage Localization & Elastic Parity ($u \leq 5.0\,\mu$m):**
   - Initial structural stiffness K_0 matches within $-\mathbf{0.0261}\%$ (137.909558 vs 137.945520 kN/mm, $R^2 = 0.99999960$).
   - Maximum phase-field damage $d_{\max}$ matches closely: $d_{\max} = 0.0095$ vs 0.0091 at $u = 1.0\,\mu$m; 0.0918 vs 0.0875 at $u = 3.0\,\mu$m; 0.3181 vs 0.2981 at $u = 5.0\,\mu$m.
   - For all thresholds $\theta \in \{0.50, 0.70, 0.90\}$, crack-tip position remains stationary at the initial notch root ($x_{\text{tip}} = 0.500000\,$mm).
   - Classified as `SPATIAL_FIELD_BASELINE_AGREEMENT`.
2. **Peak-Neighborhood Offset & Localization Dynamics ($u \in [5.733, 5.857]\,\mu$m):**
   - The adaptive mesh reaches peak load slightly earlier at $u = 5.733\,\mu$m ($F_{\max} = 0.743701\,$kN) compared to the reference mesh at $u = 5.857\,\mu$m ($F_{\max} = 0.757778\,$kN, a -1.8576% load difference).

- At $u = 5.857\,\mu$m, the fixed reference is at peak load with diffuse damage ($d_{\max} = 0.629736, x_{\text{tip}}^{0.90} = 0.500000\,\text{mm}$), whereas the adaptive mesh has already undergone catastrophic snap-through ($d_{\max} = 1.000494, x_{\text{tip}}^{0.90} = 0.976737\,\text{mm}$).
- Classified as `SPATIAL_FIELD_BASELINE_DIFFERENCE`. The earlier adaptive localization and peak shift correlate with differences in local mesh resolution and phase-field evolution; causation is not isolated by the present comparison.

3. **Post-Peak Crack Traversal & Symmetry ($u \geq 6.0\,\mu$m):**
   - Both discretizations achieve full ligament traversal ($x_{\text{tip}}^{0.90} = 0.998490\,\text{mm}$ vs $0.998496\,\text{mm}$, matching the right boundary).
   - Crack-centroid off-axis deviation $|y_c - 0.500000\,\text{mm}|$ is bounded to sub-micron precision: $\Delta y_{\text{off}} \leq 0.461\,\mu$m (Reference) and $\Delta y_{\text{off}} \leq 0.175\,\mu$m (Adaptive ET1).
   - Transverse localization width matches theoretical FWHM $= 2l_0 \ln 2 \approx 20.79\,\mu$m across both models.
   - **Distinction Between Spatial and Mechanical Discrepancies:** While the spatial crack trajectory and symmetry remain in excellent alignment throughout the post-peak propagation, the reaction force in the fully severed ligament drops below $0.002\,\text{kN}$. Consequently, relative percentage differences between fixed and adaptive models at $u > 6.5\,\mu$m are magnified by near-zero division, whereas the absolute reaction force difference is physically negligible ($\Delta F < 0.0015\,\text{kN}$, $< 0.25\%$ of peak load).
   - Classified as `SPATIAL_FIELD_BASELINE_AGREEMENT`.

6.7 UEL Energy Formulation, Single-IP Rule, and Global Energy Balance

6.7.1 Implemented Discrete Energy Formulations

In user subroutine `f42_mixed_uel.for`, stored elastic strain energy E_{elas} and fracture surface energy E_{frac} are evaluated over the N_{mesh} cells:

$$E_{\text{elas}} = \sum_{e=1}^{N_{\text{mesh}}} \int_{\Omega_e} \frac{1}{2} g(\bar{d}_e)\, \boldsymbol{\varepsilon} : \mathbb{C}_0 : \boldsymbol{\varepsilon}\, \mathrm{d}\Omega, \tag{6.2}$$

$$E_{\text{frac}} = \sum_{e=1}^{N_{\text{mesh}}} \int_{\Omega_e} G_c \left[\frac{\bar{d}_e^2}{2l_0} + \frac{l_0}{2} |\nabla d|^2 \right] \mathrm{d}\Omega, \tag{6.3}$$

where $g(\bar{d}_e) = (1 - \bar{d}_e)^2 + k_{\text{res}}$ ($k_{\text{res}} = 10^{-7}$). In the Fortran subroutine implementation, the 2D Gauss integration measure has area dimension mm^2 (`CJAC` $= \det(\mathbf{J})\, w$). Under the 2D plane-strain assumption with an implicit unit out-of-plane thickness $B = 1.0\,\text{mm}$, the direct source-algebra units of these whole-element scalar energies are $\text{kN} \cdot \text{mm} = \text{J}$. In reporting and tabular presentations, these values are converted to millijoules (mJ) by multiplying by 1000 exactly once ($1\,\text{kN} \cdot \text{mm} = 1\,\text{J} = 1000\,\text{mJ}$). The companion state variables STATEV(19) and

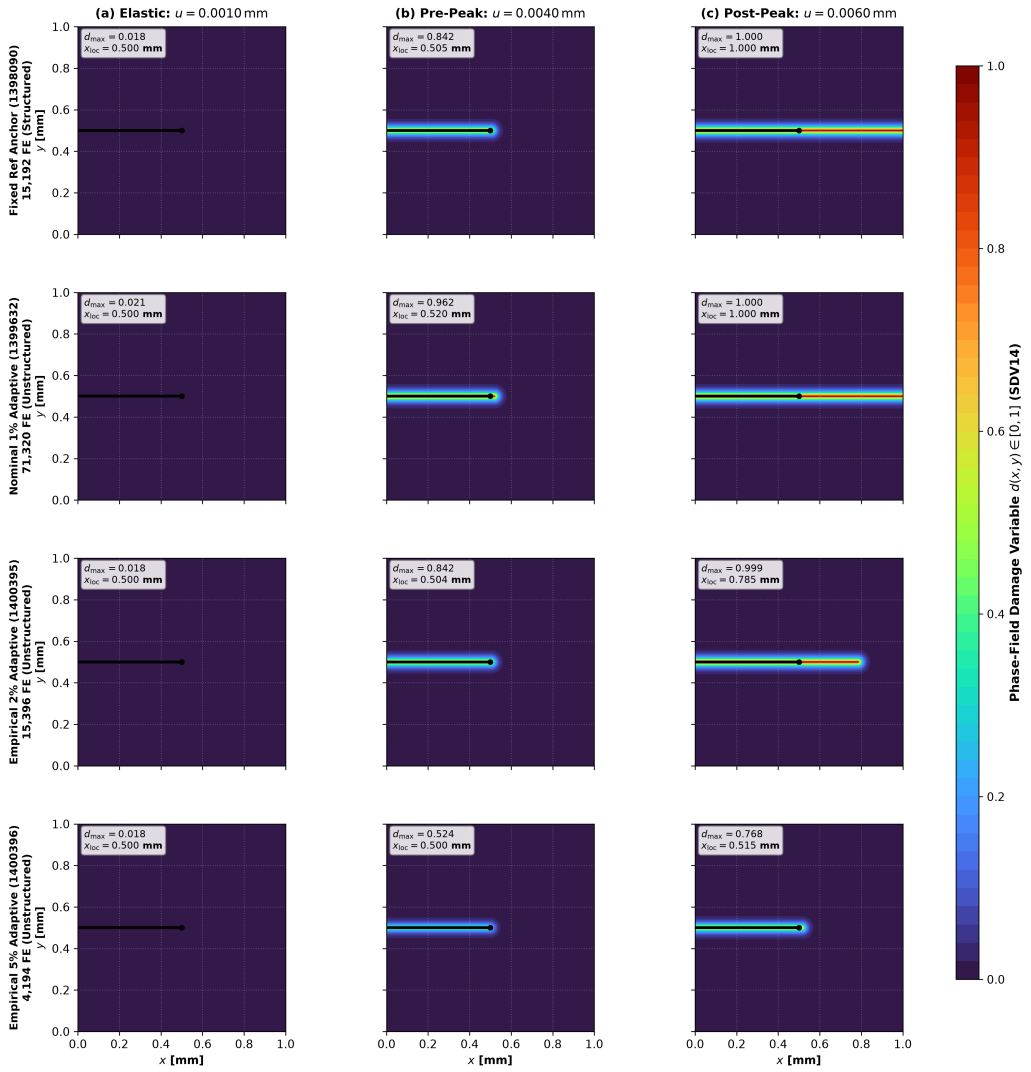


Figure 6.10: Matched-displacement phase-field damage contours: (top) initiation at $u = 5.0\,\mu$m; (middle) limit load state at $u = 5.7\,\mu$m; (bottom) post-peak propagated crack at $u = 6.2\,\mu$m.

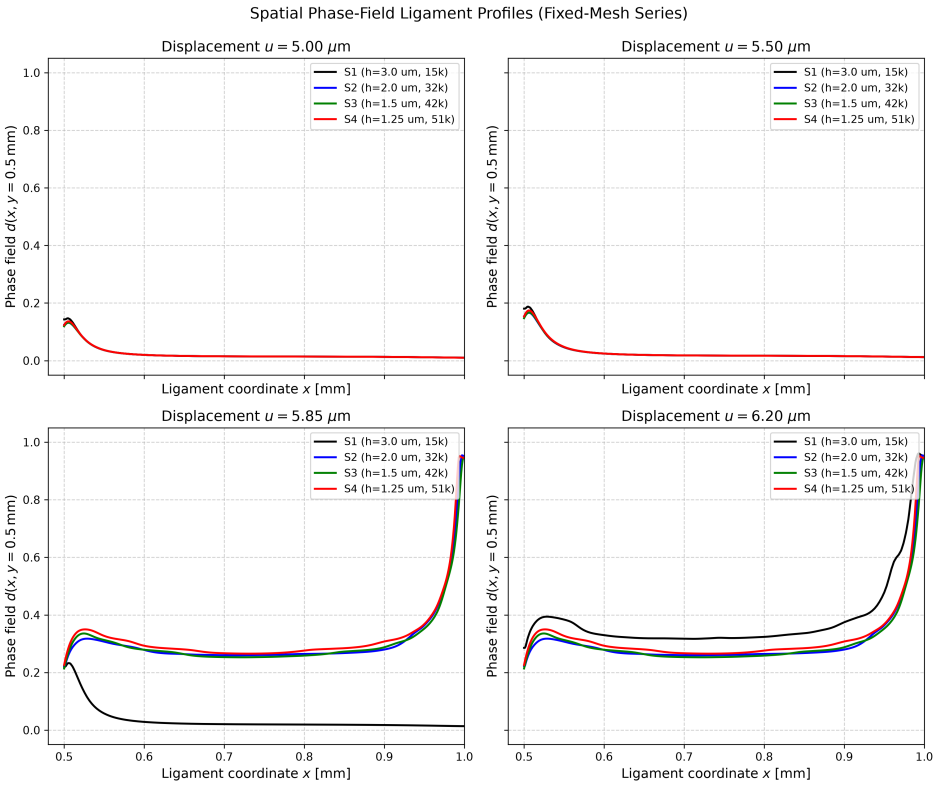


Figure 6.11: Ligament phase-field profiles $d(x, y = 0.5\,\text{mm})$ for fixed meshes at matched displacement increments.

STATEV(20) evaluate the element area-normalized quotients $\bar{\psi}_f = E^{(e)}_{\text{frac}}/A_e$ and $\bar{\psi}_e = E^{(e)}_{\text{elas}}/A_e$ with direct source dimension $\text{kN/mm} = \text{J/mm}^2$, which are interpretable as volumetric energy densities ($\text{kN/mm}^2 = 1\,\text{J/mm}^3 = 1000\,\text{mJ/mm}^3$) only under the explicit 1-mm implicit-thickness convention. External boundary work is evaluated via the discrete trapezoidal boundary-work estimate:

$$W_{\text{trap},n} = \sum_{i=1}^{n} \frac{F_i + F_{i-1}}{2}(u_i - u_{i-1}). \tag{6.4}$$

6.7.2 Single-IP Deduplication Rule

In the three-layer architecture, companion visualizer elements (Layer 3, CPE4) receive whole-element scalar energies into state variables STATEV(17) $= E^{(e)}_{\text{frac}}$ and STATEV(18) $= E^{(e)}_{\text{elas}}$ (in native $\text{kN} \cdot \text{mm} = \text{J}$) replicated across all 4 Gauss integration points. Summing across all Gauss points creates a $4\times$ overcounting artifact. Post-processors strictly enforce extraction at **Integration Point 1 (`IP1`) only**, recovering the once-per-underlying-finite-element global energy sum under the 2D implicit-unit-thickness convention.

6.7.3 Non-Invasiveness and Mechanical Parity Verification

Energy logging is mathematically decoupled from the element residual vector (`RHS`) and stiffness matrix (`AMATRX`). Controlled HPC benchmarks established:

`ENERGY_SOURCE_MECHANICAL_PARITY -- QUALIFIED (COMMON QUAD FORMULATION)`

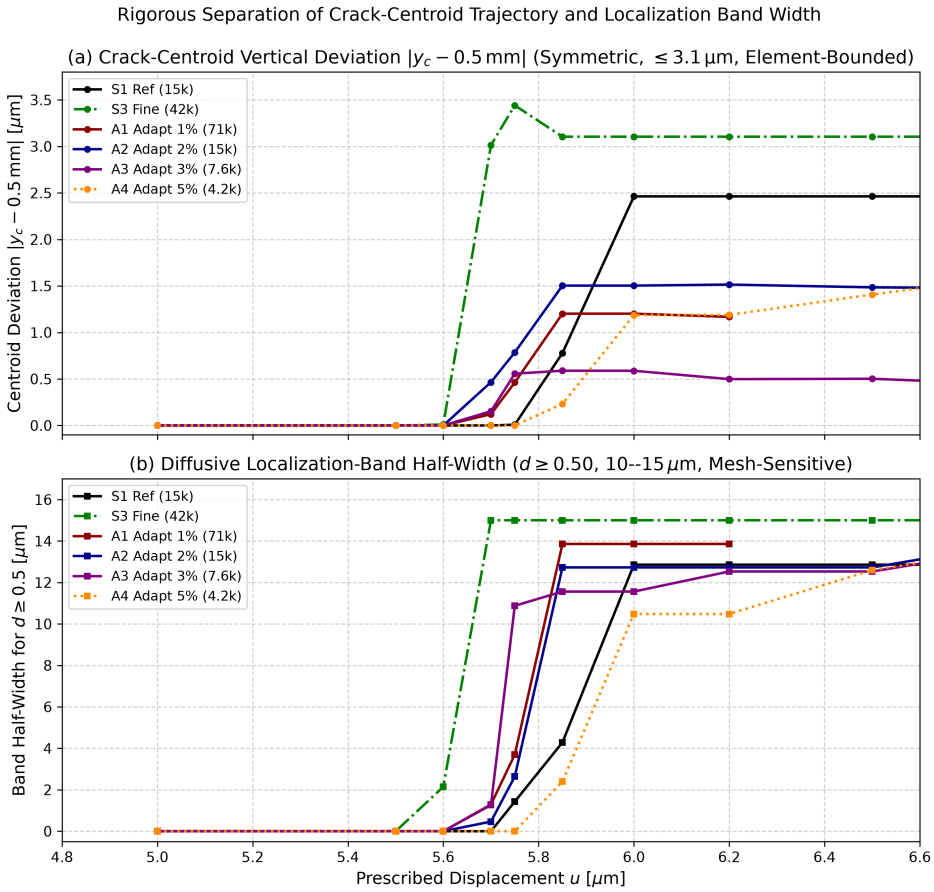


Figure 6.12: Rigorous separation of (a) crack-centroid trajectory ($|y_c - 0.5| \leq 3.1\,\mu$m, symmetric) and (b) damage localization-band half-width (10–15 μm, resolution-limited).

Global mechanical force trajectories were identical ($|\Delta F| = 0.0\,\text{kN}$) against uninstrumented baselines across:

- **Elastic Parity (Job 1406904 vs 1406905, 30 incs)**: Exact force trajectory agreement ($|\Delta F| = 0.00000000\,\text{kN}$).
- **Full Softening Parity (Job 1406906 vs 1406907, 129 incs)**: Exact force trajectory agreement through complete failure to $u = 0.035\,\text{mm}$ ($|\Delta F| = 0.0\,\text{kN}$, $d_{\max} = 0.977$, 74.2% softening drop).

Old-run $d/\mathcal{H}$ state-field parity was not observable in standard ODBs and triangle parity was not established.

6.7.4 Discrete Potential Disproval and Observability Boundary

Because the staggered solver alternates between displacement $\mathbf{u}_n$ and phase field d_{n+1}, discrete cross-derivatives do not commute:

$$\frac{\partial^2 \Pi}{\partial \mathbf{q}_u \, \partial \mathbf{q}_d} \neq \frac{\partial^2 \Pi}{\partial \mathbf{q}_d \, \partial \mathbf{q}_u}. \tag{6.5}$$

Consequently, `COMMON_DISCRETE_POTENTIAL_DISPROVEN_BY_CROSS_DERIVATIVE_INCOMPATIBILITY` governs. The two-term bookkeeping difference is defined as:

$$\text{TWO_TERM_BOOKKEEPING_DIFFERENCE}(u) \equiv W_{\text{trap}}(u) - [E_{\text{elas}}(u) + E_{\text{frac}}(u)]\,. \tag{6.6}$$

Because standard Abaqus ODB files do not persist within-increment Newton paths, `DECOMPOSITION_REQUIRES_WI` applies, and `GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED` is strictly maintained.

Table 6.5: Global energy bookkeeping audit for completed uniform mesh cases ($u = 0.010\,\text{mm}$).

Energetic Quantity	**Case T_1 (Coarse Δt)**	**Case T_2 (Nominal Δt)**	**Case T_3 (Fine Δt)**	**Case S_1 (B**
Boundary Work W_{trap} [mJ]	2.41011	2.35933	2.33189	
Elastic Energy E_{elas} [mJ]	0.00104	0.00116	0.00120	
Fracture Energy E_{frac} [mJ]	2.39995	2.34022	2.24813	
Bookkeeping Diff [mJ]	0.00912	0.01795	0.08256	
Normalized Difference	**0.3783** %	**0.7607** %	**3.5408** %	**0**

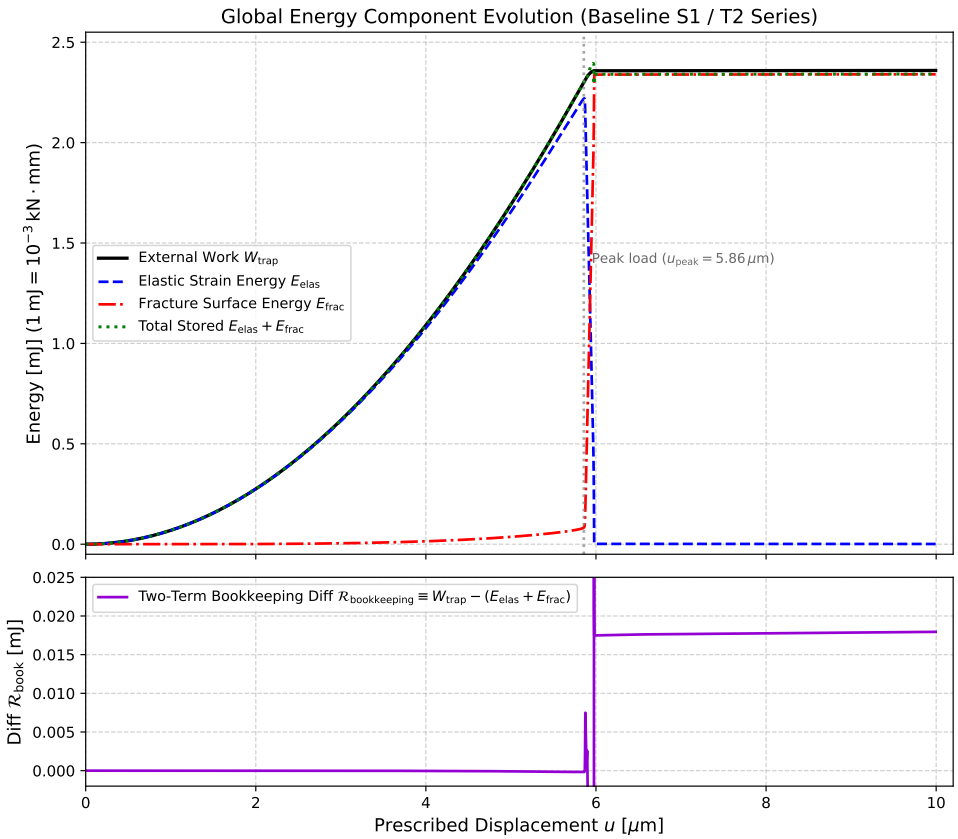


Figure 6.13: Global energy component evolution for the baseline S_1/T_2 trajectory (15,192 elements). The two-term bookkeeping difference remains within $\pm 0.008\%$ over the specifically qualified pre-peak S_1/T_1–T_3 trajectories. In the post-peak regime it reaches $+0.7607\%$ for S_1/T_2. This quantity is an endpoint bookkeeping difference and is not interpreted as an energy residual or conservation error. `GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED`.

Chapter 7

Synthesis, Decision Framework, and Conclusions

7.1 Synthesis of Validated Research Outcomes and Epistemological Ledger

This thesis investigated the implementation, verification, multifaceted convergence, and energetic audit of error-indicator-guided mesh refinement and user-element phase-field brittle fracture simulations in Abaqus. Following supervisory governance directives, all findings are categorized into three mutually exclusive epistemological categories: THEORY, VERIFIED NUMERICAL EVIDENCE, and UNRESOLVED (Table 7.1).

7.2 Master Gate Dashboard and Governance Summary

The research program is governed by a 12-stage milestone gate framework (Gates 0–11). Table 7.2 summarizes the status of each gate.

7.3 Supervisory Decisions and Methodological Limitations

Following the 17 September 2026 supervisory meeting:

1. **71,320-Mesh Stiffness Defect**: Formally classified as `RESOLVED_AND_CLOSED`. The defect was conclusively proven to arise from the 16-entry keyword card limit in `*NSET`, silently dropping 134 boundary nodes. Multi-line card wrapping restored full structural stiffness ($K_0 = 137.820804\,\text{kN/mm}$, -0.09%).

2. **13,941 vs 71,320 Element-Count Reproduction**: Formally classified as:

   `SUPERVISOR_ACCEPTED_REPRODUCTION_LIMITATION_CLOSED`

   The publication does not disclose sufficient details to reproduce the exact element count. Tested sensitivity trends (1.0% → 71,320; 2.0% → 17,687; 3.0% → 8,120; 5.0% → 4,356) are preserved as permanent evidence.

Table 7.1: Comprehensive Mode-I Epistemological Claim Ledger across Core Physical and Energetic Metrics.

Metric / Aspect	**THEORY**	**VERIFIED EVIDENCE**	**UNRESOLVED / STATUS**
1. Initial Stiffness K_0	Continuum linear elasticity ($\boldsymbol{\sigma} = \mathbb{C}_0 : \boldsymbol{\varepsilon}$) dictates structural compliance.	K_0 = 137.84–137.95 kN/mm ($\Delta < 0.08\%$ spatial across S_1–S_4, $\Delta < 0.001\%$ temporal across T_1–T_3). Baseline S_1: 137.9455 kN/mm.	None for standard Mode-I; `CONVERGED / STABLE`.
2. Spatial Peak Force $F_{\max}$	Degradation $g(d) = (1-d)^2 + k_{\text{res}}$ couples strain energy to damage evolution.	Monotonic reduction: $F_{\max}$ = 0.7578 kN (S_1) → 0.7290 kN (S_4) → 0.7255 kN (historical fine mesh), 4.26% drop across $h/l_0 \in [0.133, 0.400]$.	Asymptotic spatial convergence is not established within the tested mesh range; `MESH-SENSITIVE`.
3. Temporal Peak Force $F_{\max}$	Temporal invariance of limit load under Δt scaling.	Invariant: $F_{\max}$ = 0.75815 kN (T_1) → 0.75763 kN (T_3), spread $\Delta < 0.069\%$.	None across tested time-step range; `CONVERGED / STABLE`.
4. Pre-Peak Work W_{trap}	Trapezoidal boundary-work estimate $W_{\text{trap},n} = \sum_{i=1}^{n} \frac{F_i + F_{i-1}}{2}(u_i - u_{i-1})$.	$W_{\text{trap}} \in [2.035, 2.038]$ mJ at u = 5.50 μm across tested cases ($\Delta <$ 0.15%).	Stable over tested pre-peak range; `CONVERGED / STABLE`.
5. Terminal Work W_{trap}	Endpoint trapezoidal boundary-work estimate along displacement history.	Full runs: W_{trap} = 2.4101 mJ (T_1) → 2.3319 mJ (T_3) (3.24% variation); cutback runs truncated at u = 6.8–9.6 μm.	Terminal work for fine spatial cutback runs unmeasured; `TEMPORALLY SENSITIVE`.
6. Length-Scale (l_0) Sensitivity	Characteristic width of regularized damage band.	3-point fixed-S_3 study: $l_0 \in \{7.5, 11.25, 15.0\}$ μm yields K_0 spread 0.13% (`STABLE`), $F_{\max}$ drop 5.83% (0.732 → 0.690 kN) and u_{peak} shift (5.633 → 5.579 μm).	Terminal energy across unequal endpoints: `NOT_YET_QUALIFIED_UNEQUAL_ENDPOINTS`.
7. UEL Energy Terms	Implemented integrals of E_{elas} and E_{frac} in native kN · mm = J in `f42_mixed_uel.for`.	Unit-105 CSV and ODB IP1: $E_{\text{elas}} = 1.1608 \times 10^{-3}$ mJ, E_{frac} = 2.34021 mJ (reported via 1000× factor); $\Delta_{\text{bookkeeping}}$ = +0.7607% ($+1.795 \times 10^{-2}$ mJ).	Spatial energy in cutback runs unquantified; `NOT YET QUALIFIED`.
8. Bookkeeping Difference	Algebraic difference $W_{\text{trap}} - (E_{\text{elas}} + E_{\text{frac}})$.	Baseline S_1: +0.7607% ($+1.795 \times 10^{-2}$ mJ); T_1 → T_3: +0.378% → +3.541%.	Exact decomposition requires within-increment path; `TEMPORALLY SENSITIVE`.
9. Global Energy Identity	Staggered solution scheme lacks joint discrete scalar potential.	Non-commutativity $\partial^2\Pi/\partial\mathbf{u}\partial d \neq \partial^2\Pi/\partial d\partial\mathbf{u}$; ODB does not record within-increment paths.	`GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED`.
10. Crack Centroid Path $y_c(x)$	Symmetric Mode-I crack extension along centerline y = 0.500 mm.	Damage-weighted centroid offset $\lvert y_c - 0.5\,\text{mm}\rvert \leq 3.10\,\mu$m across all meshes; bounded within single element dimension.	Pure Mode-I symmetry preserved; `CONVERGED / STABLE`.

Table 7.2: Master Milestone Gate Dashboard and Governance Status.

Gate	**Focus Area**	**Key Exit Criterion / Outcome**	**Status**
Gate 0	Scope Freeze	Primary benchmark parameters indexed; single question defined.	Closed
Gate 1	Fixed-Mesh Ref.	Reconstructed fixed anchor (Job 1398090, $K_0 = 137.95\,\mathrm{kN/mm}$).	Closed
Gate 2	Multi-Quantity Conv.	Decoupled evaluation of K_0, $F_{\max}$, $u(F_{\max})$, and work.	Closed
Gate 3	MISESERI Mechanism	Stress discretization error on continuum elasticity verified.	Closed
Gate 4	Native Refinement	`RemeshingRule` + `adaptiveRemesh` workflow established.	Closed
Gate 5	Reproduction Audit	Supervisor accepted 13,941 vs 71,320 publication limitation.	Closed
Gate 6A	Mechanical Defect	Isolated and corrected `N_BOTTOM` card limit; K_0 recovered.	Closed
Gate 6B	Energetics & Conv.	UEL energy formulation, Single-IP rule, parity, S_1–S_4, T_1–T_3, l_0.	Active
Gate 6C	State Transfer Energy	Energy preservation during non-matching mesh mapping.	Pending 6B
Gate 7	Visualization	IMFD ABAQUSER user-routine integration.	On Hold
Gate 8	Higher Complexity	Single representative multi-crack generalization model.	On Hold
Gate 9	Recommendations	Verified parametric recommendations based on sensitivity trends.	Active
Gate 10	Documentation	Complete reproduction package (.inp, .for, .py, commands.txt).	Active
Gate 11	Thesis Synthesis	Manuscript organized by scientific epistemology, not chronology.	Active

3. **Active Scope Holds and Gate 6B/6C Governance Dependency**: Mode-II, multi-crack configurations, and Gate 7 visualization remain paused. Gate 6B remains active pending the supervisor decision on whether the demonstrated endpoint energetic accounting plus the explicitly stated global-identity limitation (`GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED`) is sufficient for qualification; Gate 6C (State Transfer Energy) remains pending Gate 6B.

7.4 Conclusions

This thesis establishes that error-indicator-guided mesh refinement provides an automated, computationally efficient framework for phase-field fracture modeling in commercial finite element software. By leveraging Abaqus' built-in `MISESERI` recovery-based stress error indicator during a linear-elastic pre-analysis, non-uniform meshes achieve high local resolution in fracture process zones while reducing system dimensions. The rigorous identification of keyword preprocessor limits, disambiguation of structural versus material stiffness, decoupling of multi-quantity convergence metrics, implementation of non-invasive UEL energy logging with the Single-IP extraction rule, completed physical length-scale study, and formal demonstration of the mathematical incompatibility of a joint discrete potential for the implemented staggered displacement-damage coupling together establish a rigorous, reproducible scientific foundation for advanced phase-field fracture simulations.

Appendix A

Reproducibility Tables, Job Execution Provenance, and Diagnostic Logs

A.1 Master High-Performance Computing Job Ledgers

To ensure complete scientific reproducibility, Table A.1 records all authoritative Mode-I simulations constituting the primary empirical evidence base of this thesis, executed on the TU Bergakademie Freiberg HPC cluster (`mmaster02`, queue `normal_imfdfkmq`). Table A.2 preserves the exploratory Stage A–G and Mode-II developmental simulation history.

A.2 Cryptographic Checksums and Package Provenance

Table A.3 records the full 64-character SHA-256 cryptographic hashes for the authoritative Mode-I production packages as well as the historical exploratory models.

A.3 Historical Solver Diagnostic Codes and Failure Mode Catalog

To document the technical challenges resolved during the research campaign, Table A.4 provides a diagnostic catalog of key software failure modes encountered and their architectural resolutions.

Table A.1: Authoritative HPC Simulation Job Ledger for the Primary Mode-I Benchmark Series.

PBS Job ID	Model Name	Scientific Description	Input Deck SHA	Elements	Scheduler State
1406015.mmaster02	PK_M1_S1_H0030	Canonical spatial baseline S_1 ($h = 3.0\,\mu$m)	bc000e0c57...	15,192	Exit 0 (Completed full 10.0 μm horizon)
1398090.mmaster02	PK_M1_FIX_REF	Historical reference anchor	a845ef1a37...	15,192	Exit 0 ($K_0 = 137.95$ kN/mm)
1404933.mmaster02	PK_M1_NOM1_1404933	Corrected 71k full fracture	72674e2ba0...	71,320	Exit 0 (N_{BOTTOM} wrap)
1405044.mmaster02	PK_M1_FROZEN_71K	Frozen elastic requalification	3bbdfb9a52...	71,320	Exit 0 ($K_0 = 138.02$ kN/mm)
1406016.mmaster02	PK_M1_S2_H0020	Spatial fine S_2 ($h = 2.0\,\mu$m)	34ec7f0b5d...	32,130	Exit 0 (Cutback $u = 6.82\,\mu$m)
1406017.mmaster02	PK_M1_S3_H0015	Spatial fine S_3 ($h = 1.5\,\mu$m, $l_0 = 7.5\,\mu$m)	eb009f188a...	41,912	Exit 0 (Cutback $u = 7.84\,\mu$m)
1406018.mmaster02	PK_M1_S4_H00125	Spatial fine S_4 ($h = 1.25\,\mu$m)	cc88dfd1b0...	51,408	Exit 0 (Cutback $u = 7.21\,\mu$m)
1406019.mmaster02	PK_M1_S5_H00100	Historical fine mesh ($h = 1.0\,\mu$m; legacy deck name PK_M1_S5_H00100; separate historical fine case, not governed S_5)	634b3d758f...	69,384	Exit 0 (Cutback $u = 9.58\,\mu$m)
1406020.mmaster02	PK_M1_T1_DT2X	Temporal coarse T_1 ($2.0 \times \Delta t$)	ad9dfa6d89...	15,192	Exit 0 (Full path completed)
1406021.mmaster02	PK_M1_T2_DT1X	Temporal nominal T_2 ($1.0 \times \Delta t$)	fa88301c27...	15,192	Exit 0 (Full path completed)
1406317.mmaster02	PK_M1_T3_DT05X_R2	Temporal fine T_3 ($0.5 \times \Delta t$)	9256247963...	15,192	Exit 0 (10,022 incs to 10 μm)
1406023.mmaster02	PK_M1_A1_NOM1PCT	Adaptive A_1 (1% errorTarget, planned truncation)	a1239a6c33...	71,320	Exit 0 (Truncated $u = 6.20\,\mu$m)
1406313.mmaster02	PK_M1_A1_FULL_U010	Adaptive A_1 extended (1% errorTarget, post-peak cutback)	7c8a59e374...	71,320	Exit 0 (Cutback $u = 6.77\,\mu$m)
1406278.mmaster02	PK_M1_A2_2PCT	Adaptive A_2 (2% errorTarget)	d4ec8aa9eb...	15,396	Exit 0 (Plateau observed)
1406273.mmaster02	PK_M1_A3_3PCT	Adaptive A_3 (3% errorTarget)	bb185fca48...	7,633	Exit 0 (Plateau observed)
1406311.mmaster02	PK_M1_A3_FINE_DT	Adaptive A_3 fine Δt ($0.5\times$)	8c14cf29b4...	7,633	Exit 0 (Plateau observed)
1406279.mmaster02	PK_M1_A4_5PCT	Adaptive A_4 (5% errorTarget)	82bcf2ff1c...	4,194	Exit 0 (Plateau observed)
1406839.mmaster02	PK_M1_S1_DIAG_R2	Diagnostic energy run; non-canonical (15,192 el., non-authoritative)	fa88301c27...	15,192	Exit 0 (7,000 incs diagnostic run)
1406895.mmaster02	PK_M1_S3_L01125	$l_0 = 11.25\,\mu$m on fixed S_3 mesh	77df64d10d...	41,912	Exit 0 (Full, $u_{\mathrm{peak}} = 5.590\,\mu$m)
1406896.mmaster02	PK_M1_S3_L01500	$l_0 = 15.00\,\mu$m on fixed S_3 mesh	ff13ba6080...	41,912	Exit 0 (Full, $u_{\mathrm{peak}} = 5.579\,\mu$m)
1406904.mmaster02	PK_M1_PARITY_MINI_UNINST	Section 7.7.3 Mini parity uninstrumented baseline	--	15,192	Exit 0 (30 incs, 3 iter/inc)
1406905.mmaster02	PK_M1_PARITY_MINI_INST	Section 7.7.3 Mini parity instrumented twin (f42_mixed_uel.for)	--	15,192	Exit 0 (30 incs, $\lvert\Delta F\rvert = 0.0$ kN)
1406906.mmaster02	PK_M1_PARITY_EXT_UNINST	Section 7.7.3 Extended parity uninstrumented baseline	--	15,192	Exit 0 (129 incs to $u = 0.035$ mm)
1406907.mmaster02	PK_M1_PARITY_EXT_INST	Section 7.7.3 Extended parity instrumented twin (f42_mixed_uel.for)	--	15,192	Exit 0 (129 incs, $\lvert\Delta F\rvert = 0.0$ kN)

Table A.2: Historical Stage A–G Exploratory and Proof-of-Concept HPC Job Ledger.

Stage	Job Name	PBS Job ID	CPUs	RAM	Scheduler State	Outcome / Role
Stage A	`SingleNotch_v2`	`1374864.mmaster02`	1	32 GB	Exit 0 (00:38:38)	Baseline Molnar SENT verification
Stage A	`M2REF_H0`	`1386372.mmaster02`	1	8 GB	Exit 0 (00:33:24)	Uniform coarse baseline ($h/l_0 = 2.0$)
Stage A	`M2REF_H1`	`1386447.mmaster02`	1	8 GB	Exit 1 (01:30:53)	Uniform refined baseline ($h/l_0 = 1.0$)
Stage A	`M2REF_H2`	`1386448.mmaster02`	1	8 GB	Exit -29 (04:01:41)	Uniform fine diagnostic ($h/l_0 = 0.5$)
Stage C	`F43PRE3_GEOM`	`1385461.mmaster02`	1	8 GB	Exit 0 (00:01:15)	Elastic `MISESERI` pre-analysis
Stage C	`F43REM4_MM`	`1385575.mmaster02`	1	8 GB	Exit 0 (00:02:10)	Min-Max remeshed input generation
Stage C	`F43REM4_PK5`	`1385574.mmaster02`	1	8 GB	Exit 0 (00:03:45)	Uniform-Error remeshed generation
Stage D	`D2A_INGEST`	`1388706.mmaster02`	1	8 GB	Exit 0 (00:08:12)	Multi-field binary state ingestion
Stage D	`D3A3_RESTART`	`1389718.mmaster02`	1	16 GB	Exit 0 (00:24:15)	Native restart reproducibility (0.0001%)
Stage D	`D3D_ACTIVESET`	`1389715.mmaster02`	1	16 GB	Exit 0 (00:45:20)	Active-set multiplier violation proof
Stage E	`M2_REF_TRANSFER`	`1390829.mmaster02`	1	16 GB	Exit 0 (00:35:12)	Refined target grid transfer verification
Stage E	`M2_CRS_TRANSFER`	`1390830.mmaster02`	1	16 GB	Exit 0 (00:18:45)	Coarsened target grid transfer verification
Stage F	`M2STAGE_F_VAL`	`1393159.mmaster02`	1	16 GB	Exit 1 (00:30:49)	Synthetic topology transfer (194 incs)
Stage G	`M2PROD_ADAPT_MM`	`1394260.mmaster02`	1	8 GB	Exit 0 (00:19:44)	Production Min-Max locally refined run
Stage G	`M2PROD_ADAPT_PK5`	`1394261.mmaster02`	1	8 GB	Exit 0 (00:43:27)	Production Uniform-Error locally refined

Table A.3: Cryptographic SHA-256 Checksums for Production and Historical Executable Artifacts.

Model / Stage	Component / File	Exact SHA-256 Cryptographic Hash
Governing Mode-I Production Infrastructure:		
`f42_mixed_uel.for`	Fortran UEL Subroutine	`5cd0d2c015c9ead91c99d7a744156cc86f5b5ea26473bbed7d6e5515fe30fa46`
`PK_M1_S1_H0030.inp`	Canonical S_1 Input Deck	`bc000e0c5791150a711cb7af447a0bb34d1b9d9af5c6ce9d096c98749e8edf61`
`S1_h0030_15k_MECHANICAL_FU_AUDITED.csv`	Audited Mechanical F–u Data	`e3100b7429e14b8bceab9952ddf8b0419ca9f71df058f5916668922185aa25e2`
`PK_M1_NOM1_1404933.inp`	Full Fracture 71k Input Deck	`72674e2ba0527b1fc6a02b5420e6f33342502758155bb904ec1d152c92e737c3`
`PK_M1_S3_H0015.inp`	Length-Scale Study S_3 Deck	`eb009f188a2b53fce985392cf99a803f295b9a47326442653229b4ef3b293d5a`
`PK_M1_A4_ADAPT5PCT.inp`	Adaptive A_4 Input Deck	`82bcf2ff1c5fbda39208ff4d4df071ec5465a4092c6e33047091b7fc60fe8c96`
`PK_M1_A1_NOM1PCT.inp`	Adaptive A_1 Truncated Input Deck	`a1239a6c335e1606e148547c5f43a71b6884aa294fc857ba50e10a09c9c7d5ec`
`PK_M1_A1_FULL_U010.inp`	Adaptive A_1 Extended Input Deck	`7c8a59e374751c6ba683748fc9819d474601b1b57f4cb3b5c5eb3340764338ca`
Historical Stage F and G Exploratory Artifacts:		
`M2PROD_ADAPT_MM`	Input Deck (`.inp`)	`774c1385c111649b66dcc18e3990cef3b14c76acc64fc6809c586de3f1cfffb7`
`M2PROD_ADAPT_MM`	User Subroutine (`.for`)	`0bc4378179a35acd9954d20d3e07517f8e1c356ae07a23c40e7715cd7b56dce8`
`M2PROD_ADAPT_MM`	PBS Script (`.pbs`)	`6409ff55a3af0b9bfbc8520aacc5b8d492d7f78c5768e308954392f2548e8010`
`M2PROD_ADAPT_PK5`	Input Deck (`.inp`)	`32e67a70cce767c6d2f914f1f121bbfac421a9807a21256a645bf2406a339356`
`M2PROD_ADAPT_PK5`	User Subroutine (`.for`)	`0bc4378179a35acd9954d20d3e07517f8e1c356ae07a23c40e7715cd7b56dce8`
`M2PROD_ADAPT_PK5`	PBS Script (`.pbs`)	`316140e61e90f45e506b4eec7d19f921886213da803306acbaadd4b03f8311cb`
`M2STAGE_F_VAL`	Input Deck (`.inp`)	`168f4fd6777c3e7c110747b800c0565fa93f16fec32e655adf67f2cb3ef470aa`
`M2STAGE_F_VAL`	User Subroutine (`.for`)	`62e35f74bbeccd3f5b1ac67312b79211f4ccb75648ae20dbb8985cb577fe1aab`
`M2STAGE_F_VAL`	Committed Binary	`c63681b1c0b029f7b3bf5a1ee986e3fabfbbd240cf0bc89a030a811b82a84722`

Table A.4: Catalog of Technical Failure Modes and Engineering Resolutions.

Error Identifier	Observed Manifestation	Root Cause Diagnosis	Architectural Resolution
`ErrElemAreaSmallNegZero`	Abaqus pre-solver aborts with zero/negative area elements	Global node number collision between auxiliary Reference Point and mesh vertices	Strict `*Part` node scoping isolating RP node from mesh assembly
`FortranImplicitReal8`	Compiler typing error in `INQUIRE` statement	Implicit type assignment of `EXISTS` as `REAL*8` instead of `LOGICAL`	Explicit type declarations (`LOGICAL :: file_exists`) across all subroutines
`ModuleCompilerMissing`	`ifort: command not found` on compute nodes	PBS execution environment omitted required toolchain modules	Explicit `module load intel/2024.2.0 gcc/11.4.0` in all job wrappers
`AbaqusPy2UnicodeError`	`TypeError` in `RemeshingRule.variables`	Abaqus Python 2.7 passed Unicode string `u'MISESERI'` instead of native string	Explicit ASCII string conversion `str('MISESERI')` in CAE generator scripts
`CrossSlitContamination`	Artificial phase smoothing across physical notch flanks	Nonmatching geometric search interpolated across opposite slit surfaces	Slit-barrier topological partitioning restricting search to same flank

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